

Gravitation Theory *II*

Homework n°1.

Claudio Pérez

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1. Hodge dual

Problem 1.1. Consider a p -form α and it's Hodge dual $\star\alpha$ in d dimensions, we must show that in the coordinate basis the expression is

$$\star\alpha = \frac{\sqrt{|g|}}{p!(d-p)!} \alpha^{\mu_1 \dots \mu_p} \epsilon_{\mu_1 \dots \mu_p \nu_1 \dots \nu_{d-p}} dx^{\nu_1} \wedge \dots \wedge dx^{\nu_{d-p}}. \quad (1)$$

To show this expression we start with the definition of the p -form α in the orthonormal basis

$$\alpha = \frac{1}{p!} \alpha_{a_1 \dots a_p} e^{a_1} \wedge \dots \wedge e^{a_p}, \quad (2)$$

and it's Hodge dual in this basis is

$$\star\alpha = \frac{1}{p!(d-p)!} \alpha^{a_1 \dots a_p} \epsilon_{a_1 \dots a_p b_1 \dots b_{d-p}} e^{b_1} \wedge \dots \wedge e^{b_{d-p}}. \quad (3)$$

We can perform a change of basis considering a rewrtng of this expression ¹

$$\begin{aligned} \star\alpha &= \frac{1}{p!(d-p)!} \alpha^{a_1 \dots a_p} \epsilon_{a_1 \dots a_p b_1 \dots b_{d-p}} e^{b_1} \wedge \dots \wedge e^{b_{d-p}} \\ &= \frac{1}{p!(d-p)!} e_{\mu_1}^{a_1} \dots e_{\mu_p}^{a_p} \alpha^{\mu_1 \dots \mu_p} \epsilon_{a_1 \dots a_p b_1 \dots b_{d-p}} e_{\nu_1}^{b_1} \dots e_{\nu_{d-p}}^{b_{d-p}} dx^{\nu_1} \wedge \dots \wedge dx^{\nu_{d-p}}, \end{aligned} \quad (4)$$

where we just have change the basis due the fact that $e^a = e_{\mu}^a dx^{\mu}$. Now we can directly identify that

$$\epsilon_{a_1 \dots a_p b_1 \dots b_{d-p}} e_{\mu_1}^{a_1} \dots e_{\mu_p}^{a_p} e_{\nu_1}^{b_1} \dots e_{\nu_{d-p}}^{b_{d-p}} = \det(e) \epsilon_{\mu_1 \dots \mu_p \nu_1 \dots \nu_{d-p}}. \quad (5)$$

Also we consider that

$$\begin{aligned} \eta_{ab} e_{\mu}^a e_{\nu}^b &= g_{\mu\nu} \quad / \det() \\ \det(g) &= \det(\eta) \det(e_{\mu}^a)^2 \\ \det(e) &= \sqrt{-g}. \end{aligned} \quad (6)$$

Then, this relations into the Hodge dual left us with

$$\begin{aligned} \star\alpha &= \frac{1}{p!(d-p)!} \alpha^{\mu_1 \dots \mu_p} \epsilon_{\mu_1 \dots \mu_p \nu_1 \dots \nu_{d-p}} \det(e) dx^{\nu_1} \wedge \dots \wedge dx^{\nu_{d-p}} \\ &= \frac{\sqrt{-g}}{p!(d-p)!} \epsilon_{\mu_1 \dots \mu_p \nu_1 \dots \nu_{d-p}} \alpha^{\mu_1 \dots \mu_p} dx^{\nu_1} \wedge \dots \wedge dx^{\nu_{d-p}}. \end{aligned} \quad (7)$$

2. Differential forms in $\mathbb{R}^3$

Problem 2.1. In $\mathbb{R}^3$ consider the euclidean metric δ_{ab} and the basis

$$e^1 = dx \quad e^2 = dy \quad e^3 = dz.$$

For $A(x^i) \in \Lambda^1(\mathbb{R}^3)$ a 1-form and $\Phi(x^i) \in \Lambda^0(\mathbb{R}^3)$ a 0-form. Compute and relate with the usual vector calculus operations in 3-dimensions the following expressions

- $\star d \star A$.
- $\star dA$.
- $\star d \star d\Phi$.

¹The notation for Levi-Civita simbol and tensor will be differenced by the use of ϵ_{abcd} and ε_{abcd} respectively.

2.1. Section a) $\star d(\star A)$

For the first expression $\star d \star A$ we start considering the expression for the 1-form

$$A = A_a e^a,$$

where $a = 1, 2, 3$ and we will use the explicit base at the end of the calculation. The Hodge dual maps $\star : \Lambda^p(M) \rightarrow \Lambda^{d-p}(M)$ so for this 1-form we have that

$$\star A = \frac{1}{2!} A^a \epsilon_{abc} e^b \wedge e^c, \quad (8)$$

is a 3-form. Here we can always rise and lower the indices with the euclidean metric if we want to do it. Next the exterior derivative maps $d : \Lambda^p(M) \rightarrow \Lambda^{p+1}(M)$ so we will have a 3-form, which is the top form of this space

$$d(\star A) = \frac{1}{2} d(A^a \epsilon_{abc} e^b \wedge e^c) = \frac{1}{2} \partial_f A^a \epsilon_{abc} e^f \wedge e^b \wedge e^c = \frac{1}{2} \partial_f A^a \epsilon_{abc} \epsilon_{fbc} Vol, \quad (9)$$

where we have used the fact that in $\mathbb{R}^3$ the relation $e^b \wedge e^c \wedge e^a = \epsilon^{abc} Vol$ is fulfilled, with Vol the volume form of $\mathbb{R}^3$ ². Finally we apply again the Hodge dual operator

$$\star d(\star A) = \star \left(\frac{1}{2} \partial_f A^a \epsilon_{abc} \epsilon_{fbc} Vol \right) = \star \left(\frac{1}{2} \partial_f A^a 2 \delta_{ab} Vol \right) = \star (\partial_a A^a Vol), \quad (10)$$

we remember that the volume form in this case is just $e^1 \wedge e^2 \wedge e^3$. The Hodge dual of the volume form in Euclidean 3-dimensional space is just 1, so we have

$$\star d(\star A) = \partial_a A_a \star (Vol) = \partial_a A_a. \quad (11)$$

Expanding the index we have explicitly

$$\star d(\star A) = \partial_x A_x + \partial_y A_y + \partial_z A_z = \vec{\nabla} \cdot \vec{A}. \quad (12)$$

2.2. Section b) $\star dA$

We follow the same procedure as before and we start computing the exterior derivative over the 1-form

$$dA = d(A_a e^a) = \partial_b A_a e^a \wedge e^b. \quad (13)$$

We can write this to show a useful property of this expression

$$\begin{aligned} \partial_b A_a e^a \wedge e^b &= \frac{1}{2} \left(\partial_b A_a e^a \wedge e^b + \partial_a A_b e^a \wedge e^b \right) \\ &= \frac{1}{2} (\partial_a A_b - \partial_b A_a) e^a \wedge e^b \\ &\equiv \frac{1}{2} F_{ab} e^a \wedge e^b, \end{aligned} \quad (14)$$

with F_{ab} antisymmetric. Now we compute the dual

$$\star \left(\frac{1}{2} F_{ab} e^a \wedge e^b \right) = \frac{1}{2} F^{ab} \epsilon_{abc} e^c = \frac{1}{2} (\partial_a A_b - \partial_b A_a) \epsilon_{abc} e^c, \quad (15)$$

²Remember that in the case of $\mathbb{R}^3$ we have that $\epsilon_{abc} = \epsilon^{abc}$ because we raise indices with δ_{ab} .

which is a 1-form. Because the antisymmetry of the Levi-Civita symbol in a and b , we see that the expression reduces to

$$\star(dA) = \frac{1}{2}\epsilon_{abc}\partial_a A_b e^c. \quad (16)$$

To see what is the vector calculus expression we replace explicitly the basis and obtain

$$\epsilon_{ab1}\partial_a A_b dx = (\epsilon_{231}\partial_2 A_3 + \epsilon_{321}\partial_3 A_2) dx = (\partial_y A_z - \partial_z A_y) dx, \quad (17)$$

$$\epsilon_{ab2}\partial_a A_b dy = (\epsilon_{312}\partial_3 A_1 + \epsilon_{132}\partial_1 A_3) dy = (\partial_z A_x - \partial_x A_z) dy, \quad (18)$$

$$\epsilon_{ab3}\partial_a A_b dz = (\epsilon_{123}\partial_1 A_2 + \epsilon_{213}\partial_2 A_1) dz = (\partial_x A_y - \partial_y A_x) dz, \quad (19)$$

and this expression is exactly the expression of $\vec{\nabla} \times \vec{A}$, then we conclude

$$\star(dA) \longleftrightarrow \vec{\nabla} \times \vec{A}. \quad (20)$$

2.3. Section c) $\star d \star d\Phi$

The same procedure as before lead us to start with the exterior derivative of the 0-form Φ . By definition we will have

$$d\Phi = \partial_a \phi e^a, \quad (21)$$

and this is 1-form. The Hodge dual is

$$\star(d\Phi) = \frac{1}{2!}\partial_a \Phi \epsilon_{abc} e^b \wedge e^c, \quad (22)$$

which is a 2-form. The exterior derivative of this expression is

$$d(\star d\Phi) = \frac{1}{2}\partial_d \partial_a \Phi \epsilon_{abc} e^d \wedge e^b \wedge e^c = \frac{1}{2}\partial_d \partial_a \Phi \epsilon_{abc} \epsilon_{dbc} Vol, \quad (23)$$

where we have used again the fact that the volume element can be related to the *top-form*, and this term is a 3-form. Then this term is simply

$$d(\star d\Phi) = \frac{1}{2}\partial^a \partial_a \Phi Vol. \quad (24)$$

Until now all is consistent with the mapping that defines the operations that we have used

$$\Lambda^0(\mathbb{R}^3) \xrightarrow{d} \Lambda^1(\mathbb{R}^3) \xrightarrow{\star} \Lambda^2(\mathbb{R}^3) \xrightarrow{d} \Lambda^3(\mathbb{R}^3).$$

Now we apply the Hodge dual again to obtain the expression that we want

$$\star d \star d\Phi = \frac{1}{2}\delta^{ab}\partial_b \partial_a \Phi \star Vol = \delta^{ab}\partial_b \partial_a \phi. \quad (25)$$

Expanding in the basis we have that

$$\delta^{ab}\partial_b \partial_a \Phi = \partial_x^2 \Phi + \partial_y^2 \Phi + \partial_z^2 \Phi, \quad (26)$$

and this expression is just $\nabla^2 \Phi$ in the vector calculus. Then we conclude

$$\star d \star d\Phi \longleftrightarrow \nabla^2 \Phi.$$

3. Maxwell in Differential Forms

Problem 3.1. *Formulate the Maxwell electrodynamics in differential forms language. Consider that*

$$A = A_a e^a, \quad (27)$$

$$J = J_a e^a, \quad (28)$$

$$F = dA. \quad (29)$$

With this determine

- The action principle.
- The field equations obtained by the variation of the action and the Bianchi identities. Write this results in a coordinate basis and obtain the Maxwell equations in the usual vector notation.
- The energy-momentum obtained from the variation of the action. Write this results in the coordinate basis.
- The wave equation for the electromagnetic potential.

3.1. Section a) Maxwell action with source

We want to determine the action principle of Maxwell theory coupled to an external source in four dimensions, in the language of differential forms. The only expression that we assume as previously known is the tensorial action

$$S = \int d^4x \sqrt{-g} \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + A_\mu J^\mu \right), \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (30)$$

and our goal is to show that the differential forms objects $F \wedge \star F$ and $A \wedge \star J$ reproduce exactly the two terms of this expression.

First we set the definitions to understand where our objects are living. The gauge potential is an *ordinary* differential form: the gauge group $U(1)$ has a one-dimensional Lie algebra with a single generator that commutes with itself, so the algebra factor is trivial and we can simply write

$$A = A_a e^a \in \Omega^1(M), \quad J = J_a e^a \in \Omega^1(M), \quad (31)$$

where the source J must also be a 1-form: this is forced by the structure of the coupling, since $\star J$ is then a 3-form and $A \wedge \star J \in \Omega^4(M)$ is a top form, ready to be integrated. The curvature is defined as

$$F = dA \in \Omega^2(M). \quad (32)$$

Note that there is no $A \wedge A$ term here: for an ordinary 1-form the coefficients are commuting numbers, so

$$A \wedge A = A_a A_b e^a \wedge e^b = \frac{1}{2} (A_a A_b - A_b A_a) e^a \wedge e^b = 0, \quad (33)$$

where we antisymmetrized using $e^b \wedge e^a = -e^a \wedge e^b$ and the fact that $A_a A_b = A_b A_a$. The self-interaction term is identically zero.

Now we obtain the explicit form of the curvature. Since the only contribution is the exterior derivative

$$\begin{aligned} F &= dA = d(A_b e^b) = \partial_a A_b e^a \wedge e^b \\ &= \frac{1}{2} (\partial_a A_b - \partial_b A_a) e^a \wedge e^b \\ &= \frac{1}{2} F_{ab} e^a \wedge e^b, \end{aligned} \quad (34)$$

with components $F_{ab} = \partial_a A_b - \partial_b A_a$, which in the coordinate basis $e^a \rightarrow dx^\mu$ reduce to the tensorial expression $F_{\mu\nu}$ of (30).

$F \wedge \star F$ equals the kinetic Lagrangian:

The Hodge dual of the curvature 2-form is also a 2-form in 4-dimensions

$$\begin{aligned} \star F &= \star \frac{1}{2} F_{ab} e^a \wedge e^b \\ &= \frac{1}{2} \frac{1}{(4-2)!} F^{ab} \epsilon_{abcd} e^c \wedge e^d. \end{aligned} \quad (35)$$

Then we can construct the 4-form

$$\begin{aligned} F \wedge \star F &= \frac{1}{2} F_{ab} e^a \wedge e^b \wedge \frac{1}{4} F^{cd} \epsilon_{cdef} e^e \wedge e^f \\ &= \frac{1}{8} F_{ab} F^{cd} \epsilon_{cdef} e^a \wedge e^b \wedge e^e \wedge e^f \\ &= \frac{1}{8} F_{ab} F^{cd} \epsilon_{cdef} \epsilon^{abef} \det(e) d^4x, \end{aligned} \quad (36)$$

where we used the basic identity $e^a \wedge e^b \wedge e^e \wedge e^f = \epsilon^{abef} \det(e) d^4x$. The contraction of two Levi-Civita symbols with two contracted indices is

$$\epsilon^{abef} \epsilon_{cdef} = 2 \left(\delta_c^a \delta_d^b - \delta_d^a \delta_c^b \right), \quad (37)$$

so we have

$$\begin{aligned} F \wedge \star F &= \frac{1}{8} \cdot 2 \left(F_{ab} F^{ab} - F_{ab} F^{ba} \right) \det(e) d^4x \\ &= \frac{1}{2} F_{ab} F^{ab} \det(e) d^4x, \end{aligned} \quad (38)$$

where we used the antisymmetry $F^{ba} = -F^{ab}$. There is no trace required here: since the algebra factor is trivial, $F \wedge \star F$ is already an ordinary scalar 4-form.

$A \wedge \star J$ equals the source coupling:

The Hodge dual of the source 1-form is a 3-form

$$\star J = \frac{1}{(4-1)!} J^a \epsilon_{abcd} e^b \wedge e^c \wedge e^d, \quad (39)$$

and the coupling 4-form is

$$\begin{aligned} A \wedge \star J &= A_f e^f \wedge \frac{1}{6} J^a \epsilon_{abcd} e^b \wedge e^c \wedge e^d \\ &= \frac{1}{6} A_f J^a \epsilon_{abcd} e^f \wedge e^b \wedge e^c \wedge e^d \\ &= \frac{1}{6} A_f J^a \epsilon_{abcd} \epsilon^{fbcd} \det(e) d^4x. \end{aligned} \quad (40)$$

Here we need the contraction of two Levi-Civita symbols with three contracted indices

$$\epsilon^{fbcd}\epsilon_{abcd} = 3! \delta_a^f = 6 \delta_a^f, \quad (41)$$

which gives directly

$$A \wedge \star J = \frac{1}{6} \cdot 6 A_a J^a \det(e) d^4x = A_a J^a \det(e) d^4x. \quad (42)$$

Finally, the flat scalars coincide with the curved ones, $F_{ab}F^{ab} = F_{\mu\nu}F^{\mu\nu}$ and $A_a J^a = A_\mu J^\mu$, because the pairs of inverse vielbeins contracted with η reconstruct the inverse metric $g^{\mu\rho} = \eta^{ac}e_a^\mu e_c^\rho$, and $\det(e) = \sqrt{-g}$. Therefore, comparing with (30), we identify the Maxwell action with source in the language of differential forms

$$S = \int \left(-\frac{1}{2} F \wedge \star F + A \wedge \star J \right) = \int d^4x \det(e) \left(-\frac{1}{4} F_{ab}F^{ab} + A_a J^a \right) \quad (43)$$

with the global factor $-\frac{1}{2}$ of the kinetic term fixed precisely so that the tensor normalization $-\frac{1}{4}$ is recovered, while the source term enters with coefficient one.

3.2. Section b) Equations of Motion

With the action principle formulated in differential forms language, we can use the variational principle to obtain the dynamics of the gauge field. Starting from

$$S[A] = \int \left(-\frac{1}{2} F \wedge \star F + A \wedge \star J \right), \quad (44)$$

where the source J is external and therefore not varied, we apply the variation directly

$$\begin{aligned} \delta S[A] &= -\frac{1}{2} \int (\delta F \wedge \star F + F \wedge \star \delta F) + \int \delta A \wedge \star J \\ &= - \int \delta F \wedge \star F + \int \delta A \wedge \star J, \end{aligned} \quad (45)$$

where we used the fact that for ordinary p -forms $\alpha \wedge \star \beta = \beta \wedge \star \alpha$, so the two kinetic terms are equal. Now we expand the curvature. Since $F = dA$ and the variation commutes with the exterior derivative³

$$\delta F = \delta(dA) = d(\delta A). \quad (46)$$

This is the great simplification of the abelian theory: the variation of the curvature is an exact form, with no extra terms. Using the Leibniz rule for the wedge product

$$d(\delta A \wedge \star F) = d(\delta A) \wedge \star F + (-1)^1 \delta A \wedge d\star F, \quad (47)$$

we can write

$$\delta S[A] = - \int d(\delta A \wedge \star F) - \int \delta A \wedge d\star F + \int \delta A \wedge \star J. \quad (48)$$

The first term is a total derivative: proposing appropriate boundary conditions, $\delta A|_{\partial M} = 0$, and using the Stokes theorem, that term vanishes [section A](#). Then

$$\delta S[A] = \int \delta A \wedge (-d\star F + \star J), \quad (49)$$

³The variation is defined through a one-parameter family $A_s = A + s\eta$, so $\delta(dA) = \frac{d}{ds}\big|_{s=0} d(A + s\eta) = d\eta = d(\delta A)$: δ acts on the configuration space while d acts on spacetime, and derivatives with respect to independent variables commute [1].

and in order to have an extremum, i.e., $\delta S = 0$ for all δA , we obtain the equation of motion

$$\boxed{d\star F = \star J} . \quad (50)$$

The source appears exactly where it should: as the inhomogeneity of the field equation. The second set of equations comes for free from the definition of the curvature. Since $F = dA$ and the exterior derivative is nilpotent, $d^2 = 0$, we have the Bianchi identity

$$\boxed{dF = d^2 A = 0} , \quad (51)$$

which is an identity, valid for any configuration, unaffected by the presence of the source.

There is an important consistency condition hidden in the field equation. Applying d to both sides and using the nilpotency of the exterior derivative

$$d^2 \star F = 0 \implies d\star J = 0 , \quad (52)$$

the source cannot be arbitrary: it must satisfy $d\star J = 0$, which in components is the continuity equation $\nabla_\mu J^\mu = 0$. Current conservation is not an extra postulate but a consistency requirement of the coupled theory.

Coordinate basis

Now we write the equations in the coordinate basis. We expand

$$A = A_\mu dx^\mu , \quad F = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu , \quad J = J_\mu dx^\mu , \quad (53)$$

and the Hodge duals, using the coordinate basis expression obtained before (7), have components

$$(\star F)_{\rho\sigma} = \frac{\sqrt{-g}}{2} \epsilon_{\alpha\beta\rho\sigma} F^{\alpha\beta} , \quad \star J = \frac{\sqrt{-g}}{3!} J^\alpha \epsilon_{\alpha\nu\rho\sigma} dx^\nu \wedge dx^\rho \wedge dx^\sigma . \quad (54)$$

Since the coordinate basis is holonomic, $d(dx^\rho) = 0$, the exterior derivative acts only on the components

$$d\star F = \frac{1}{2} \partial_\nu (\star F)_{\rho\sigma} dx^\nu \wedge dx^\rho \wedge dx^\sigma . \quad (55)$$

Both sides of the equation of motion are 3-forms in 4-dimensions, which have the same content as their dual 1-forms, so we contract both with $\epsilon^{\mu\nu\rho\sigma}$. For the left hand side, replacing the components of $\star F$

$$\begin{aligned} \epsilon^{\mu\nu\rho\sigma} \cdot \frac{1}{2} \partial_\nu (\star F)_{\rho\sigma} &= \frac{1}{4} \epsilon^{\mu\nu\rho\sigma} \epsilon_{\alpha\beta\rho\sigma} \partial_\nu (\sqrt{-g} F^{\alpha\beta}) \\ &= \frac{1}{4} \cdot 2 (\delta_\alpha^\mu \delta_\beta^\nu - \delta_\beta^\mu \delta_\alpha^\nu) \partial_\nu (\sqrt{-g} F^{\alpha\beta}) \\ &= \partial_\nu (\sqrt{-g} F^{\mu\nu}) , \end{aligned} \quad (56)$$

using the contraction (37) and the antisymmetry of F . For the right hand side, the contraction with three contracted indices (41) gives

$$\epsilon^{\mu\nu\rho\sigma} \cdot \frac{\sqrt{-g}}{6} \epsilon_{\alpha\nu\rho\sigma} J^\alpha = \frac{1}{6} \cdot 6 \delta_\alpha^\mu \sqrt{-g} J^\alpha = \sqrt{-g} J^\mu . \quad (57)$$

Equating both sides, the equation of motion in the coordinate basis is

$$\boxed{\frac{1}{\sqrt{-g}} \partial_\nu (\sqrt{-g} F^{\mu\nu}) = J^\mu} , \quad (58)$$

which in flat spacetime reduces to $\partial_\nu F^{\mu\nu} = J^\mu$, the inhomogeneous Maxwell equations with source. For the Bianchi identity, the components of $dF = 0$ are the totally antisymmetric derivative

$$\boxed{\partial_\mu F_{\nu\rho} + \partial_\nu F_{\rho\mu} + \partial_\rho F_{\mu\nu} = 0}, \quad (59)$$

which are the homogeneous Maxwell equations, and the consistency condition $d \star J = 0$ becomes, by the same dualization as above,

$$\frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} J^\mu) = 0, \quad (60)$$

the covariant continuity equation. The full set of Maxwell equations is recovered.

Recovering the vector Maxwell equations

To make contact with the usual vector language, we work in flat spacetime with $\eta_{ab} = \text{diag}(-1, +1, +1, +1)$ and identify the components of the field strength with the electric and magnetic fields

$$E_i = F_{i0}, \quad B_i = \frac{1}{2} \epsilon_{ijk} F_{jk} \iff F_{ij} = \epsilon_{ijk} B_k, \quad (61)$$

with $i, j, k = 1, 2, 3$, while the source decomposes as $J^\mu = (\rho, \vec{J})$. The temporal component $\mu = 0$ of the equation of motion $\partial_\nu F^{\mu\nu} = J^\mu$ gives

$$\partial_i F^{0i} = \partial_i E_i = \rho \implies \nabla \cdot \vec{E} = \rho, \quad (62)$$

where we used $F^{0i} = g^{00} g^{ij} F_{0j} = -F_{0i} = F_{i0} = E_i$. This is the Gauss law. The spatial components $\mu = i$ give

$$\partial_0 F^{i0} + \partial_j F^{ij} = -\partial_t E_i + \epsilon_{ijk} \partial_j B_k = J^i \implies \nabla \times \vec{B} - \partial_t \vec{E} = \vec{J}, \quad (63)$$

the Ampère–Maxwell law. The Bianchi identity $\partial_{[\mu} F_{\nu\rho]} = 0$ provides the other two equations. Taking all indices spatial, $(\mu\nu\rho) = (ijk)$, and contracting with ϵ_{ijk}

$$\epsilon_{ijk} \partial_i F_{jk} = \epsilon_{ijk} \epsilon_{jkl} \partial_i B_l = 2 \partial_i B_i = 0 \implies \nabla \cdot \vec{B} = 0, \quad (64)$$

the absence of magnetic monopoles, and taking the mixed components $(0ij)$

$$\partial_0 F_{ij} + \partial_i F_{j0} + \partial_j F_{0i} = \epsilon_{ijk} \partial_t B_k + \partial_i E_j - \partial_j E_i = 0 \implies \nabla \times \vec{E} + \partial_t \vec{B} = 0, \quad (65)$$

the Faraday law. The four Maxwell equations are recovered, cleanly split by their geometric origin: the two inhomogeneous ones come from the variational principle $d\star F = \star J$, and the two homogeneous ones from the nilpotency $dF = d^2 A = 0$.

3.3. Section c) Energy-Momentum tensor

From the Maxwell action formulated in the orthonormal frame (43), the energy-momentum of the electromagnetic field is obtained from the kinetic term

$$S_{\text{kin}} = -\frac{1}{4} \int d^4x \det(e) F_{ab} F^{ab}, \quad (66)$$

since the source term $A \wedge \star J$ describes the interaction with external matter, and its contribution to the energy-momentum balance is conventionally assigned to the matter sector.

We can see what happens if instead of a metric we have vielbeins to describe the space-time. The dependence on the vielbein is more transparent writing

$$F_{ab} = e_a^\mu e_b^\nu F_{\mu\nu}, \quad (67)$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is independent of the vielbein, as is the flat metric η^{ab} that raises the indices of F^{ab} . Taking the variation of the action

$$\delta_e S_{\text{kin}} = -\frac{1}{4} \int d^4x \left[\delta(\det(e)) F_{ab} F^{ab} + \det(e) \delta(F_{ab} F^{ab}) \right]. \quad (68)$$

The variation of the determinant follows from $\delta \ln \det M = \text{Tr}(M^{-1} \delta M)$

$$\delta e = e e_c^\lambda \delta e_\lambda^c, \quad (69)$$

and from the fundamental relation $e_a^\mu e_\nu^a = \delta_\nu^\mu$, the variation of the inverse vielbein is

$$\delta e_a^\mu = -e_b^\mu e_a^\nu \delta e_\nu^b. \quad (70)$$

For the kinetic scalar, keeping $F_{\mu\nu}$ fixed

$$\begin{aligned} \delta(F_{ab} F^{ab}) &= 2 \delta F_{ab} F^{ab} = 2 \delta(e_a^\mu e_b^\nu F_{\mu\nu}) F^{ab} \\ &= 2(\delta e_a^\mu e_b^\nu + e_a^\mu \delta e_b^\nu) F_{\mu\nu} F^{ab} \\ &= 4 \delta e_a^\mu (e_b^\nu F_{\mu\nu} F^{ab}) \\ &= -4 e_a^\lambda F_{cb} F^{ab} \delta e_\lambda^c, \end{aligned} \quad (71)$$

where in the third line we renamed indices of the second term and used the double antisymmetry of F , and in the last line we inserted the variation of the inverse vielbein. Adding both contributions

$$\delta_e S_{\text{kin}} = -\frac{1}{4} \int d^4x \det(e) \left[e_c^\lambda F_{ab} F^{ab} - 4 e_a^\lambda F_{cb} F^{ab} \right] \delta e_\lambda^c. \quad (72)$$

Then we define the energy-momentum “tensor” as the functional derivative

$$T_c^\lambda \equiv \frac{1}{\det(e)} \frac{\delta S_{\text{kin}}}{\delta e_\lambda^c} = -\frac{1}{4} e_c^\lambda F_{ab} F^{ab} + e_a^\lambda F_{cb} F^{ab}, \quad (73)$$

a hybrid object with one flat and one curved index, inherited from the structure of the vielbein with respect to which we varied. Contracting the curved index with e^d_λ and lowering with η_{db} , after renaming $c \rightarrow a$, $d \rightarrow b$, we obtain the fully flat expression

$$T_{ab} = F_{ac} F_b^c - \frac{1}{4} \eta_{ab} F_{cd} F^{cd}. \quad (74)$$

The translation to the coordinate basis is a pure projection with vielbeins, $T_{\mu\nu} = e^a_\mu e^b_\nu T_{ab}$, where the contracted pair of inverse vielbeins reconstructs the inverse metric, giving

$$T_{\mu\nu} = F_{\mu\rho} F_\nu^\rho - \frac{1}{4} g_{\mu\nu} F_{\rho\sigma} F^{\rho\sigma}. \quad (75)$$

This is exactly the standard electromagnetic energy-momentum tensor obtained from the tensorial action by varying with respect to the metric. As consistency checks: it is symmetric; it is traceless in four dimensions, $g^{\mu\nu} T_{\mu\nu} = F_{\mu\rho} F^{\mu\rho} - \frac{4}{4} F_{\rho\sigma} F^{\rho\sigma} = 0$, reflecting the conformal invariance of Maxwell theory in $d = 4$; and its components reproduce the familiar electromagnetic energy density and Poynting flux. Moreover, in the presence of the source the tensor is no longer conserved by itself: using the equations of motion one finds $\nabla_\mu T^{\mu\nu} = F^{\nu\rho} J_\rho$, the Lorentz force density, showing that the field exchanges energy and momentum with the external matter exactly as expected.

3.4. Section d) Wave equation

The equation of motion can be turned into a dynamical equation for the potential itself. Substituting $F = dA$ into the field equation

$$d\star dA = \star J, \quad (76)$$

we obtain a second order equation for A . To extract its components we work in flat space-time, where we can choose a global inertial frame in which the orthonormal coframe is also holonomic, $e^a = dx^a$, so that ∂_a are ordinary derivatives and η_{ab} raises and lowers all indices. In this frame the equation of motion in components reads, from Section b),

$$\partial_b F^{ab} = J^a, \quad (77)$$

and inserting $F^{ab} = \partial^a A^b - \partial^b A^a$

$$\partial_b (\partial^a A^b - \partial^b A^a) = \partial^a (\partial_b A^b) - \square A^a = J^a, \quad (78)$$

where $\square \equiv \eta^{ab} \partial_a \partial_b = -\partial_t^2 + \nabla^2$ is the d'Alembertian. The equation couples all the components of A through the divergence term, but we can use the gauge symmetry to remove it. Under a gauge transformation

$$A \rightarrow A + d\lambda, \quad (79)$$

the curvature is invariant by nilpotency, $F \rightarrow d(A + d\lambda) = dA + d^2\lambda = F$, while the divergence shifts as

$$\partial_a A^a \rightarrow \partial_a A^a + \square \lambda, \quad (80)$$

so choosing λ as a solution of $\square \lambda = -\partial_a A^a$, which always exists, we can impose the *Lorenz gauge*

$$\partial_a A^a = 0 \iff d\star A = 0, \quad (81)$$

where the second expression is the same condition written in the language of differential forms: the divergence of a 1-form is the dual of the exact 4-form $d\star A$. In this gauge the first term of (78) drops and we are left with the wave equation

$$\boxed{\square A^a = -J^a}. \quad (82)$$

Each component of the potential satisfies an independent wave equation sourced by the corresponding component of the current. In vacuum, $J^a = 0$, the equation $\square A^a = 0$ describes electromagnetic waves propagating at the speed of light: the existence of light as a propagating solution is contained entirely in this expression. Note also that the Lorenz condition does not fix the gauge completely: residual transformations with $\square \lambda = 0$ preserve it, which is the remaining freedom used to reach, for example, the radiation gauge in vacuum.

3.5. Section e)Extra: Wave equation in curved background

The equation of motion can be turned into a dynamical equation again but in a curved background where the operators that appear are different depending on the manifold. Substituting $F = dA$ into the field equation

$$d\star dA = \star J, \quad (83)$$

we obtain a second order equation for A . From Section b), the components of this equation in the coordinate basis are

$$\nabla_\nu F^{\mu\nu} = J^\mu, \quad (84)$$

where ∇_μ is the Levi-Civita covariant derivative, since $\frac{1}{\sqrt{-g}}\partial_\nu(\sqrt{-g}F^{\mu\nu}) = \nabla_\nu F^{\mu\nu}$ for an antisymmetric tensor. Because the connection is torsion-free, the field strength can be written equivalently with covariant derivatives, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = \nabla_\mu A_\nu - \nabla_\nu A_\mu$ (the Christoffel symbols cancel by symmetry), so

$$\nabla_\nu (\nabla^\mu A^\nu - \nabla^\nu A^\mu) = J^\mu. \quad (85)$$

Now we have a important difference, to isolate the divergence we need to exchange the order of the covariant derivatives in the first term, and covariant derivatives do not commute. Using the Ricci identity $[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho_{\sigma\mu\nu}V^\sigma$ and contracting to the Ricci tensor

$$\nabla_\nu \nabla^\mu A^\nu = \nabla^\mu (\nabla_\nu A^\nu) + R^\mu_{\nu} A^\nu, \quad (86)$$

the equation (85) becomes

$$\nabla^\mu (\nabla_\nu A^\nu) + R^\mu_{\nu} A^\nu - \square A^\mu = J^\mu, \quad (87)$$

with $\square \equiv \nabla_\nu \nabla^\nu$ the curved d'Alembertian. As in the flat case, the divergence term is removed with the gauge symmetry: under $A \rightarrow A + d\lambda$ the curvature is invariant by nilpotency, while the divergence shifts as $\nabla_\mu A^\mu \rightarrow \nabla_\mu A^\mu + \square\lambda$, so solving the scalar wave equation $\square\lambda = -\nabla_\mu A^\mu$ we can impose the *Lorenz gauge*

$$\nabla_\mu A^\mu = 0 \quad \Longleftrightarrow \quad d\star A = 0, \quad (88)$$

which has the same expression in the language of differential forms as in flat spacetime. In this gauge we obtain the curved wave equation

$$\boxed{\square A^\mu - R^\mu_{\nu} A^\nu = -J^\mu}. \quad (89)$$

The Ricci term naturally arises here by the non-commutativity of the covariant derivatives, and it shows that the potential does not propagate freely on a curved background but couples directly to the local curvature. This combination is precisely the de Rham wave operator acting on 1-forms: in the language of differential forms, the Lorenz gauge condition together with the equation of motion give $\Delta A = (d\delta + \delta d)A = \delta F$, and the Weitzenböck identity $\Delta = -\square + \text{Ric}$ on 1-forms reproduces exactly the boxed expression [7, 8].

4. Yang-Mills in Differential Forms

Problem 4.1. *Formulate the Yang-Mills theory in the differential forms language. Consider that the gauge potential is given by*

$$A = A_a^B(x)e^a \otimes T_B,$$

with $a = 0, 1, 2, 3$ and $B = 1, \dots, \dim(\mathfrak{g})$, and considering that the Yang-Mills curvature is given by

$$F = dA + A \wedge A.$$

Determine

- *The action principle.*
- *The field equations from the variation of the action and the Bianchi identities. Write the results in a coordinate basis.*
- *The energy-momentum tensor obtained from the action. Write the results in a coordinate basis.*

4.1. Section a) Yang-Mills action

We want to determine the action principle of Yang-Mills theory in four dimensions which we can write in coordinate or orthonormal basis

$$S = -\frac{1}{4} \int d^4x \text{Tr} (F_{\mu\nu} F^{\mu\nu}) = -\frac{1}{4} \int \det(e) \text{Tr} F_{ab} F^{ab}. \quad (90)$$

First we will set the definitions to understand where are living our objects. The gauge potential A is not an ordinary differential form but a form *valued in the Lie algebra* $\mathfrak{g}$. Concretely, it lives in the tensor product space $A \in \Omega^1(M) \otimes \mathfrak{g}$, where $\Omega^1(M)$ is the space of 1-forms on the spacetime manifold M and $\mathfrak{g}$ is the Lie algebra of the gauge group. This is reflected in its expansion $A = A_a^B(x) e^a \otimes T_B$, which carries a form index a (contracted with the coframe e^a) and an algebra index B (contracted with the generators T_B). More generally, any algebra-valued p -form lives in $\Omega^p(M) \otimes \mathfrak{g}$, and the curvature, built from A through $F = dA + A \wedge A$, is consequently a 2-form valued in the algebra, $F \in \Omega^2(M) \otimes \mathfrak{g}$, with expansion $F = \frac{1}{2} F_{ab}^C e^a \wedge e^b \otimes T_C$. The wedge product on such objects is defined factor by factor, $(\alpha \otimes X) \wedge (\beta \otimes Y) = (\alpha \wedge \beta) \otimes (XY)$.

Now we can obtain directly the explicit form of the gauge curvature with the gauge field valued on the algebra. The first part is given by the exterior derivative

$$\begin{aligned} dA &= d(A_b^C e^b \otimes T_C) = \partial_a A_b^C e^a \wedge e^b \otimes T_C \\ &= \frac{1}{2} (\partial_a A_b^C - \partial_b A_a^C) e^a \wedge e^b \otimes T_C \\ &= \partial_{[a} A_{b]}^C e^a \wedge e^b \otimes T_C. \end{aligned} \quad (91)$$

The next contribution is obtained directly by replacing the same definition

$$\begin{aligned} A \wedge A &= (A_a^A e^a \otimes T_A) \wedge (A_b^B e^b \otimes T_B) = (A_a^A A_b^B e^a \wedge e^b) \otimes T_A T_B \\ &= \frac{1}{2} (A_a^A A_b^B e^a \wedge e^b \otimes T_A T_B - A_b^A A_a^B e^a \wedge e^b \otimes T_A T_B) \\ &= \frac{1}{2} A_a^A A_b^B e^a \wedge e^b \otimes T_A T_B - \frac{1}{2} A_b^B A_a^A e^a \wedge e^b \otimes T_B T_A \end{aligned} \quad (92)$$

$$= \frac{1}{2} A_a^A A_b^B e^a \wedge e^b \otimes (T_A T_B - T_B T_A), \quad (93)$$

and using that the generators satisfy the Lie algebra

$$[T_A, T_B] = f_{AB}^C T_C, \quad (94)$$

with f_{AB}^C the structure constants of the algebra, we have

$$\begin{aligned} A \wedge A &= \frac{1}{2} A_a^A A_b^B e^a \wedge e^b \otimes (T_A T_B - T_B T_A) \\ &= \frac{1}{2} A_a^A A_b^B f_{AB}^C e^a \wedge e^b \otimes T_C. \end{aligned} \quad (95)$$

Then the total expression for the curvature is given by

$$\begin{aligned} F &= dA + A \wedge A \\ &= \partial_{[a} A_{b]}^C e^a \wedge e^b \otimes T_C + \frac{1}{2} A_a^A A_b^B f_{AB}^C e^a \wedge e^b \otimes T_C \\ &= \frac{1}{2} (\partial_a A_b^C - \partial_b A_a^C + A_a^A A_b^B f_{AB}^C) e^a \wedge e^b \otimes T_C \\ &= \frac{1}{2} F_{ab}^C e^a \wedge e^b \otimes T_C. \end{aligned} \quad (96)$$

This expression can be directly converted in the usual coordinate frame expression by a change of basis $e^a \rightarrow dx^\mu$, where we obtain (see [section B](#))

$$F_{\mu\nu}^C = \partial_{[\mu} A_{\nu]} + f_{AB}^C A_\mu^A A_\nu^B. \quad (97)$$

Now, how we can build an action principle in terms of differential forms independent of a basis? We notice that the F is a two form valued on the Lie algebra. In consequence the Hodge dual is also a two form in 4-dimensions. We know

$$\begin{aligned} \star F &= \star \frac{1}{2} F_{ab}^C e^a \wedge e^b \otimes T_C \\ &= \frac{1}{2} \frac{1}{(4-2)!} \left(F^{abC} \epsilon_{abcd} e^c \wedge e^d \otimes T_C \right). \end{aligned} \quad (98)$$

Then we can construct the 4-form

$$\begin{aligned} F \wedge \star F &= \frac{1}{2} F_{ab}^C e^a \wedge e^b \otimes T_C \wedge \frac{1}{4} \left(F^{cdD} \epsilon_{cdef} e^e \wedge e^f \otimes T_D \right) \\ &= \frac{1}{2} \frac{1}{4} F_{ab}^C F^{cdD} \epsilon_{cdef} e^a \wedge e^b \wedge e^e \wedge e^f \otimes T_C T_D \\ &= \frac{1}{8} F_{ab}^C F^{cdD} \epsilon_{cdef} \left(\det(e) \epsilon^{abef} d^4 x \right) \otimes T_C T_D \\ &= \frac{1}{8} \det(e) F_{ab}^C F^{cdD} (2!(4-2)!) \delta_{[cd]}^{ab} d^4 x \otimes T_C T_D \\ &= \frac{1}{2} \det(e) F_{ab}^C F^{cdD} \frac{1}{2} \left(\delta_{cd}^{ab} - \delta_{dc}^{ab} \right) d^4 x \otimes T_C T_D \\ &= \frac{1}{4} F_{ab}^C F^{abD} \det(e) d^4 x \otimes T_C T_D. \end{aligned} \quad (99)$$

We can integrate this form in 4-dimensions, but this expression is valued on the Lie algebra, and we can notice that this still transform in the adjoint representation. To extract a gauge-invariant scalar from $F \wedge \star F \in \Omega^4(M) \otimes \mathfrak{g}$, we use the group trace, defined as the operator $\text{Tr} \equiv \text{id}_{\Omega^p} \otimes \text{tr}_{\mathfrak{g}}$, which acts as the identity on the form factor and as the ordinary matrix trace on the algebra factor [4]. On a factorizable element this means $\text{Tr}(\alpha \otimes X) = \alpha \text{tr}(X)$, so the differential form α passes through untouched while $\text{tr}(X)$ reduces to a number. With the normalization $\text{Tr}(T_C T_D) = \delta_{CD}$, the trace therefore contracts the two algebra indices of $F \wedge \star F$ against each other, turning the matrix-valued 4-form into an ordinary scalar 4-form ready to be integrated. Its gauge invariance follows from the cyclicity of the trace: under $F \mapsto g F g^{-1}$, one has $\text{Tr}(g F g^{-1} \wedge \star(g F g^{-1})) = \text{Tr}(F \wedge \star F)$. Then we construct this term

$$\begin{aligned} \text{Tr} F \wedge \star F &= \text{Tr} -\frac{1}{4} F_{ab}^C F^{abD} \det(e) d^4 x \otimes T_C T_D \\ &= -\frac{1}{4} F_{ab}^C F^{abD} \det(e) d^4 x \delta_{CD} \\ &= -\frac{1}{4} F_{ab}^C F_C^{ab} \det(e) d^4 x, \end{aligned} \quad (100)$$

which is the term to build the Yang-Mills action in the language of differential forms

$$S = -\frac{1}{4} \int \det(e) F_{ab}^C F_C^{ab} d^4 x = -\frac{1}{4} \int \det(e) \text{Tr} \left(F_{ab}^C F^{abD} \right) d^4 x. \quad (101)$$

4.2. Section b) Equations of Motion

With the formulation of this action principle in differential forms language, we can use the variational principle to obtain the dynamics of the gauge fields. Starting from

$$S[A] = -\frac{1}{4} \int \text{Tr}(F \wedge \star F), \quad (102)$$

we can apply the variation directly and obtain this term

$$\begin{aligned} \delta S[A] &= -\frac{1}{2} \int \delta \text{Tr}(F \wedge \star F) \\ &= -\frac{1}{4} \int \text{Tr}(\delta F \wedge \star F + F \wedge \delta \star F) \\ &= -\int \text{Tr}(\delta F \wedge \star F). \end{aligned} \quad (103)$$

We make use of the fact that for Lie algebra valued p -forms

$$\text{Tr}(\alpha \wedge \star \beta) = \text{Tr}(\beta \wedge \star \alpha),$$

because $\alpha^A \wedge \star \beta^B = \beta^B \wedge \star \alpha^A$ and the trace is symmetric. Now we explicitly expand the gauge curvature

$$\begin{aligned} -\int \text{Tr}(\delta F \wedge \star F) &= -\int \text{Tr}(\delta(dA + A \wedge A) \wedge \star F) \\ &= -\int \text{Tr}(\delta(dA) \wedge \star F + \delta(A \wedge A) \wedge \star F). \end{aligned} \quad (104)$$

The first term in (104) is explicitly

$$\delta dA \wedge \star F = d(\delta A) \wedge \star F, \quad (105)$$

where we can make use of the Leibniz rule for the wedge product

$$d(\delta A \wedge \star F) = d(\delta A) \wedge \star F + (-1)^1 \delta A \wedge d(\star F), \quad (106)$$

then this term is exactly

$$\text{Tr}(\delta dA \wedge \star F) = \text{Tr}(d(\delta A \wedge \star F)) + \text{Tr}(\delta A \wedge (d \star F)). \quad (107)$$

The second term in (104) is expanded as

$$\text{Tr}(\delta(A \wedge A) \wedge \star F) = \text{Tr}(\delta A \wedge A \wedge \star F) + \text{Tr}(A \wedge \delta A \wedge \star F), \quad (108)$$

here the first term is in the form we want. The other contribution can be manipulated using the fact that, for $\alpha \in \Omega^p \otimes \mathfrak{g}$ and $\beta \in \Omega^q \otimes \mathfrak{g}$

$$\text{Tr}(\alpha \wedge \beta) = (-1)^{pq} \text{Tr} \beta \wedge \alpha. \quad (109)$$

The expression that we have is $\alpha = A$ so is a 1-form, and $\delta A \wedge \star F$ is a 1-form product with a 2-form, so it is a 3-form. Then we have

$$\text{Tr}(A \wedge \delta A \wedge \star F) = (-1) \text{Tr}(\delta A \wedge \star F \wedge A). \quad (110)$$

Collecting all the terms (from (107), (108) and (110)) we have in the variation of the action

$$\delta S[A] = -\frac{1}{2} \int [\text{Tr}(d(\delta A \wedge \star F)) + \text{Tr}(\delta A \wedge d \star F) + \text{Tr}(\delta A \wedge A \wedge \star F) - \text{Tr}(\delta A \wedge \star F \wedge A)] . \quad (111)$$

We can propose appropriated boundary conditions to vanish the total derivative present in the first term such as $\delta A|_{\partial M} = 0$, and using the stokes theorem vanish that term [section A](#). Then we can have more compact this expression

$$\delta S[A] = - \int \text{Tr}[\delta A \wedge (d \star F + A \wedge \star F - \star F \wedge A)] . \quad (112)$$

Finally we can recognize this expression as

$$D \star F = d \star F + A \wedge \star F - \star F \wedge A , \quad (113)$$

which is the graded commutator

$$D \star F = d \star F + [A, \star F]_{\wedge} , \quad [A, \Phi]_{\wedge} \equiv A \wedge \Phi - (-1)^p \Phi \wedge A , \quad (114)$$

where p is the degree of the form Φ . In our case $\Phi = \star F$ is a 2-form, so $(-1)^p = (-1)^2 = +1$ and the relative minus sign between the two terms follows directly. With all of this procedure, finally our variational principle gives us

$$\delta S[A] = - \int \text{Tr}(\delta A \wedge D \star F) , \quad (115)$$

and in order to have this as a extremum, i.e, $\delta S = 0, \forall \delta A$, we have the equation of motion

$$D \star F = 0 , \quad (116)$$

expression that we can expand to see the non abelian nature of this expression

$$d \star F + A \wedge \star F - \star F \wedge A = 0 , \quad (117)$$

where we need to remember that we have differential forms valued in the algebra, so the last two terms do not commute.

Now we can write the equation of motion in the coordinate basis. We expand all the objects in the basis $\{dx^\mu\}$

$$A = A_\mu^A dx^\mu \otimes T_A , \quad F = \frac{1}{2} F_{\mu\nu}^C dx^\mu \wedge dx^\nu \otimes T_C . \quad (118)$$

The Hodge dual of the curvature 2-form is also a 2-form in 4-dimensions, and using the coordinate basis expression obtained before (7), its components are

$$\star F = \frac{1}{2} (\star F)_{\rho\sigma}^C dx^\rho \wedge dx^\sigma \otimes T_C , \quad (\star F)_{\rho\sigma}^C = \frac{\sqrt{-g}}{2} \epsilon_{\alpha\beta\rho\sigma} F^{C\alpha\beta} , \quad (119)$$

where the indices of $F^{C\alpha\beta} = g^{\alpha\mu} g^{\beta\nu} F_{\mu\nu}^C$ are raised with the metric. Since the coordinate basis is holonomic, $d(dx^\rho) = 0$, the exterior derivative acts only on the components

$$d \star F = \frac{1}{2} \partial_\nu (\star F)_{\rho\sigma}^C dx^\nu \wedge dx^\rho \wedge dx^\sigma \otimes T_C . \quad (120)$$

The next contribution is the graded commutator. Computing each piece with the tensor product rule

$$A \wedge \star F = \frac{1}{2} A_\nu^A (\star F)_{\rho\sigma}^B dx^\nu \wedge dx^\rho \wedge dx^\sigma \otimes T_A T_B , \quad (121)$$

and for the second term we move dx^ν through the two dx 's of $\star F$, collecting $(-1)^2 = +1$

$$\begin{aligned}\star F \wedge A &= \frac{1}{2}(\star F)_{\rho\sigma}^B A_\nu^A dx^\rho \wedge dx^\sigma \wedge dx^\nu \otimes T_B T_A \\ &= \frac{1}{2}A_\nu^A (\star F)_{\rho\sigma}^B dx^\nu \wedge dx^\rho \wedge dx^\sigma \otimes T_B T_A.\end{aligned}\quad (122)$$

The form parts are now identical and only the order of the generators differs, so the difference gives us the commutator of the algebra, in the same way as it occurred for $A \wedge A$

$$A \wedge \star F - \star F \wedge A = \frac{1}{2}A_\nu^A (\star F)_{\rho\sigma}^B dx^\nu \wedge dx^\rho \wedge dx^\sigma \otimes [T_A, T_B]. \quad (123)$$

Using the Lie algebra $[T_A, T_B] = f_{AB}^C T_C$ and collecting all the terms, the full 3-form is

$$\begin{aligned}D \star F &= \frac{1}{2} \left(\partial_\nu (\star F)_{\rho\sigma}^C + f_{AB}^C A_\nu^A (\star F)_{\rho\sigma}^B \right) dx^\nu \wedge dx^\rho \wedge dx^\sigma \otimes T_C \\ &= \frac{1}{2} D_\nu (\star F)_{\rho\sigma}^C dx^\nu \wedge dx^\rho \wedge dx^\sigma \otimes T_C,\end{aligned}\quad (124)$$

where we recognize in the parenthesis the gauge covariant derivative of the components. Then the equation of motion $D \star F = 0$ tell us that the totally antisymmetric part vanishes

$$D_{[\nu} (\star F)_{\rho\sigma]}^C = 0. \quad (125)$$

A 3-form in 4-dimensions has the same content as its dual 1-form, so we can contract with $\epsilon^{\mu\nu\rho\sigma}$ and replace the components of $\star F$

$$\epsilon^{\mu\nu\rho\sigma} D_\nu (\star F)_{\rho\sigma}^C = \frac{\sqrt{-g}}{2} \epsilon^{\mu\nu\rho\sigma} \epsilon_{\alpha\beta\rho\sigma} D_\nu F^{C\alpha\beta}, \quad (126)$$

and using the contraction of two Levi-Civita symbols with two contracted indices

$$\epsilon^{\mu\nu\rho\sigma} \epsilon_{\alpha\beta\rho\sigma} = 2 \left(\delta_\alpha^\mu \delta_\beta^\nu - \delta_\beta^\mu \delta_\alpha^\nu \right), \quad (127)$$

the global numerical factor and sign are irrelevant for the equation equated to zero, so we obtain

$$\boxed{\frac{1}{\sqrt{-g}} \partial_\nu \left(\sqrt{-g} F^{C\nu\mu} \right) + f_{AB}^C A_\nu^A F^{B\nu\mu} = 0} \quad (128)$$

which in flat spacetime, where $\sqrt{-g} = 1$, reduces to the standard expression

$$\boxed{D_\nu F^{C\nu\mu} \equiv \partial_\nu F^{C\nu\mu} + f_{AB}^C A_\nu^A F^{B\nu\mu} = 0}. \quad (129)$$

We can make two consistency checks of this result. First, for the abelian case $f_{AB}^C = 0$ we recover exactly $\partial_\nu F^{\nu\mu} = 0$, the sourceless Maxwell equations. Second, the term $f_{AB}^C A_\nu^A F^{B\nu\mu}$ shows that the gauge field acts as a source of itself: this is the manifestation in the equations of motion of the non-abelian nature that appeared when the order of the generators prevented the cancellation of the two terms in the graded commutator.

4.3. Section c) Energy-Momentum tensor

In the usual actions formulated in the coordinate basis tensors, we know that a gauge theory has an energy-momentum tensor coming from the matter-background interaction defined by [6]

$$T_{\mu\nu}[A, g] = -\frac{2}{\sqrt{-g}} \frac{\delta S[A, g]}{\delta g^{\mu\nu}}. \quad (130)$$

From the Yang-Mills action formulated in the orthonormal frame (101)

$$S = -\frac{1}{4} \int d^4x \det(e) \text{Tr}(F_{ab} F^{ab}) = -\frac{1}{4} \int d^4x \det(e) \text{Tr}(F_{ab} \eta^{ac} \eta^{bd} F_{cd}), \quad (131)$$

we can see what happens if instead of a metric we have vielbeins to describe the space-time. Directly taking the variation of the action

$$\begin{aligned} \delta_e S &= -\frac{1}{4} \delta \int d^4x \det(e) \text{Tr}(F_{ab} F^{ab}) \\ &= -\frac{1}{4} \int d^4x \left[\delta(\det(e)) \text{Tr}(F_{ab} F^{ab}) + \det(e) \delta \text{Tr}(F_{ab} F^{ab}) \right]. \end{aligned} \quad (132)$$

Before explicitly expanding these variations, we need to comment something about the second term. The expression F_{ab}^C is dependent on the vielbein, but we can expand it to explicitly show how it depends on the vielbein and work only with the dependent part to calculate the variation. The dependence on the vielbein is more transparent writing

$$F_{ab}^C = e_a^\mu e_b^\nu F_{\mu\nu}^C, \quad (133)$$

where $F_{\mu\nu}^C$ is independent of the vielbein, and similarly

$$F^{abC} = \eta^{ac} \eta^{bd} e_c^\mu e_d^\nu F_{\mu\nu}^C, \quad (134)$$

where the metric η^{ab} is also independent of the vielbein. Now, writing it in this way, we can compute first the variation of the determinant ⁴

$$\delta e = e e_a^\mu \delta e_\mu^a. \quad (135)$$

Now we focus on the second term. For simplicity we will omit the algebra indices in this calculation

$$\begin{aligned} \delta(F_{ab} F^{ab}) &= \delta F_{ab} F^{ab} + F_{ab} \delta F^{ab} \\ &= 2 \delta F_{ab} F^{ab} \\ &= 2 \delta (e_a^\mu e_b^\nu F_{\mu\nu}) F^{ab}. \end{aligned} \quad (136)$$

Using the fundamental relation $e_a^\mu e_\nu^\mu = \delta_a^\nu$, the variation of the inverse vielbein is given by

$$\delta e_a^\mu = -e_b^\mu e_a^\nu \delta e_\nu^b. \quad (137)$$

Then the variation (136) is

$$\begin{aligned} 2 \delta (e_a^\mu e_b^\nu F_{\mu\nu}) F^{ab} &= 2 (\delta e_a^\mu e_b^\nu F_{\mu\nu} + e_a^\mu \delta e_b^\nu F_{\mu\nu}) F^{ab} \\ &= 2 \delta e_a^\mu e_b^\nu F_{\mu\nu} F^{ab} + 2 e_b^\nu \delta e_a^\mu F_{\nu\mu} F^{ba} \\ &= 4 \delta e_a^\mu (e_b^\nu F_{\mu\nu} F^{ab}) \\ &= -4 e_c^\mu e_a^\lambda \delta e_\lambda^c (e_b^\nu F_{\mu\nu} F^{ab}) \\ &= -4 e_a^\lambda F_{cb} F^{ab} \delta e_\lambda^c. \end{aligned} \quad (138)$$

⁴We use the usual relation $\delta \ln \det M = \text{Tr}(M^{-1} \delta M)$.

We have the two contributions of the variation, so adding all we have

$$\begin{aligned}\delta_e S &= -\frac{1}{4} \int d^4x \left[\left(\det(e) e_c^\lambda \delta e_\lambda^c \right) \text{Tr} \left(F_{ab} F^{ab} \right) + \det(e) \text{Tr} \left(-4 e_a^\lambda F_{cb} F^{ab} \delta e_\lambda^c \right) \right] \\ &= -\frac{1}{4} \int d^4x \det(e) \left[e_c^\lambda F_{ab}^C F^{abC} - 4 e_a^\lambda F_{cb}^C F^{abC} \right] \delta e_\lambda^c.\end{aligned}\quad (139)$$

Then we can define a type of energy-momentum “tensor” as the functional derivative

$$T_c^\lambda \equiv \frac{1}{\det(e)} \frac{\delta S[A, e]}{\delta e_\lambda^c}, \quad (140)$$

so we identify our result from the variation of the action as

$$T_c^\lambda = -\frac{1}{4} e_c^\lambda F_{ab}^C F^{abC} + e_a^\lambda F_{cb}^C F^{abC}. \quad (141)$$

This is a hybrid object, with one flat and one curved index, inherited directly from the structure of the vielbein with respect to which we varied. We can convert the curved index to a flat one contracting with e^d_λ

$$T_c^d = T_c^\lambda e^d_\lambda = -\frac{1}{4} \delta_c^d F_{ab}^C F^{abC} + F_{cb}^C F^{dbC}, \quad (142)$$

and lowering the index d with the flat metric η_{ab} , after renaming $c \rightarrow a$, $d \rightarrow b$, we can write the final result in this compact expression

$$T_{ab} = F_{ac}^C F_b{}^{cC} - \frac{1}{4} \eta_{ab} F_{cd}^C F^{cdC} = \text{Tr} \left[F_{ac} F_b{}^c - \frac{1}{4} \eta_{ab} F_{cd} F^{cd} \right], \quad (143)$$

where in the last equality we used the normalization $\text{tr}(T_C T_D) = \delta_{CD}$ to write the sum over algebra indices as the group trace.

Translation to the coordinate basis

The translation to the coordinate basis is a pure projection with vielbeins, $T_{\mu\nu} = e^a_\mu e^b_\nu T_{ab}$. Applying it to the first term, each vielbein converts a flat index into a curved one

$$e^a_\mu e^b_\nu F_{ac}^C F_b{}^{cC} = e^a_\mu e^b_\nu F_{ac}^C \eta^{cd} F_{bd}^C = F_{\mu\rho}^C g^{\rho\sigma} F_{\nu\sigma}^C = F_{\mu\rho}^C F_\nu{}^\rho{}^C, \quad (144)$$

where the contracted pair of inverse vielbeins inside η^{cd} reconstructed the inverse spacetime metric $g^{\rho\sigma} = \eta^{cd} e^\rho_c e^\sigma_d$. For the second term we use directly $g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu$, and the scalar $F_{cd}^C F^{cdC} = F_{\rho\sigma}^C F^{C\rho\sigma}$ is basis independent. Then

$$T_{\mu\nu} = F_{\mu\rho}^C F_\nu{}^\rho{}^C - \frac{1}{4} g_{\mu\nu} F_{\rho\sigma}^C F^{C\rho\sigma}. \quad (145)$$

This is exactly the energy-momentum tensor that one obtains from the tensorial action $S = -\frac{1}{4} \int d^4x \sqrt{-g} F_{\mu\nu}^C F^{C\mu\nu}$ by varying with respect to the metric, so both procedures match. This is not a coincidence: from $g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu$ the variations are related by $\delta g_{\mu\nu} = 2 \eta_{ab} e^a_{(\mu} \delta e^b_{\nu)}$, so the vielbein variation contains the metric variation plus an antisymmetric piece. Since our result T_{ab} turned out to be symmetric, the antisymmetric piece does not contribute and both methods coincide. As consistency checks, the tensor is symmetric, it is traceless in four dimensions, $\eta^{ab} T_{ab} = F_{ab}^C F^{abC} - \frac{4}{4} F_{cd}^C F^{cdC} = 0$, reflecting the conformal invariance of Yang-Mills in $d = 4$, and in the abelian case the sum over the algebra index collapses to a single term, recovering exactly the Maxwell energy-momentum tensor.

5. Carroll Algebra

Problem 5.1. Consider the Carroll algebra defined by the commutation relations

$$\begin{aligned} [\mathbf{J}_{ab}, \mathbf{P}_c] &= 2\delta_{c[a}\mathbf{P}_{b]}, \\ [\mathbf{J}_{ab}, \mathbf{G}_c] &= 2\delta_{c[a}\mathbf{G}_{b]}, \\ [\mathbf{J}_{ab}, \mathbf{J}_{cd}] &= 4\delta_{[a[d}\mathbf{J}_{c]b]}, \\ [\mathbf{P}_a, \mathbf{G}_b] &= 2\delta_{ab}\mathbf{H}. \end{aligned}$$

In this notation the generators $\mathbf{P}_a$ are the spatial translations, the $\mathbf{J}_{ab}$ the rotations, $\mathbf{H}$ time translations and $\mathbf{G}_a$ the boosts. Also the infinitesimal gauge parameter and the gauge field are given by

$$\varepsilon = \zeta\mathbf{H} + \zeta^a\mathbf{G}_a + \lambda^a\mathbf{P}_a + \frac{1}{2}\lambda^{ab}\mathbf{J}_{ab}, \quad (146)$$

$$A_\mu = \tau_\mu\mathbf{H} + \omega_\mu^a\mathbf{G}_a + e_\mu^a\mathbf{P}_a + \frac{1}{2}\omega_\mu^{ab}\mathbf{J}_{ab}. \quad (147)$$

Determine

- The infinitesimal local gauge transformation of the fields $\tau_\mu, \omega_\mu^a, e_\mu^a$ and ω_μ^{ab} .
- The components of the curvature

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu].$$

- The form in which the curvature transforms under infinitesimal gauge transformations.

5.1. Section a) Infinitesimal transformation of A_μ

In order to determine the local gauge transformation of A_μ valued on the Lie algebra we make use of the infinitesimal form of the transformation

$$\delta A_\mu = D_\mu \epsilon = \partial_\mu \epsilon + [A_\mu, \epsilon].$$

We see directly that the partial derivatives acts on the coefficients of each generator on ϵ

$$\partial_\mu \epsilon = \partial_\mu \zeta \mathbf{H} + \partial_\mu \zeta^a \mathbf{G}_a + \partial_\mu \lambda^a \mathbf{P}_a + \frac{1}{2} \partial_\mu \lambda^{ab} \mathbf{J}_{ab}. \quad (148)$$

For the Lie bracket, we just check each contribution reading the algebra of generators, where all the combinations that doesn't are considered there are just zero. So, for the expression fully expanded we have

$$-[A_\mu, \epsilon] = \left[\zeta \mathbf{H} + \zeta^a \mathbf{G}_a + \lambda^a \mathbf{P}_a + \frac{1}{2} \lambda^{ab} \mathbf{J}_{ab}, \tau_\mu \mathbf{H} + \omega_\mu^a \mathbf{G}_a + e_\mu^a \mathbf{P}_a + \frac{1}{2} \omega_\mu^{ab} \mathbf{J}_{ab} \right]. \quad (149)$$

We can see that the contributions that gives proportional to $\mathbf{H}$ are given from

$$\left[\omega_\mu^a \mathbf{G}_a + e_\mu^a \mathbf{P}_a, \zeta^b \mathbf{G}_b + \lambda^b \mathbf{P}_b \right] = \left[\omega_\mu^a \mathbf{G}_a, \lambda^b \mathbf{P}_b \right] + \left[e_\mu^a \mathbf{P}_a, \zeta^b \mathbf{G}_b \right]. \quad (150)$$

We simply expand each one of the brackets using the properties of bi-linearity to get out the coefficients. The first bracket is

$$\left[\omega_\mu^a \mathbf{G}_a, \lambda^b \mathbf{P}_b \right] = \omega_\mu^a \lambda^b [\mathbf{G}_a, \mathbf{P}_b] = -\omega_\mu^a \lambda^b (2\delta_{ba}\mathbf{H}) = -2\omega_\mu^a \lambda_a \mathbf{H}. \quad (151)$$

The second one is

$$\left[e_\mu^a \mathbf{P}_a, \zeta^b \mathbf{G}_b \right] = e_\mu^a \zeta^b [\mathbf{P}_a, \mathbf{G}_b] = e_\mu^a \zeta^b (2\delta_{ab} \mathbf{H}) = 2e_\mu^a \zeta_a \mathbf{H}. \quad (152)$$

Recalling that the algebra tell us that the rest of the brackets does not gives another contribution to the $\mathbf{H}$ transformation, we see that the transformation is given by the components proportional to $\mathbf{H}$

$$\delta\tau_\mu \mathbf{H} = \partial_\mu \zeta \mathbf{H} + 2e_\mu^a \zeta_a \mathbf{H} - 2\omega_\mu^a \lambda_a \mathbf{H}, \quad (153)$$

in other words

$$\delta\tau_\mu = \partial_\mu \zeta + 2e_\mu^a \zeta_a - 2\omega_\mu^a \lambda_a. \quad (154)$$

The next contributions to each generator is computed in the same procedure. The terms coming from the bracket that goes with $\mathbf{G}_a$ are given by

$$\left[\omega_\mu^a \mathbf{G}_a + \frac{1}{2} \omega_\mu^{ab} \mathbf{J}_{ab}, \zeta^b \mathbf{G}_b + \frac{1}{2} \lambda^{cd} \mathbf{J}_{cd} \right] = \left[\frac{1}{2} \omega_\mu^{ab} \mathbf{J}_{ab}, \zeta^c \mathbf{G}_c \right] + \left[\omega_\mu^c \mathbf{G}_c, \frac{1}{2} \lambda^{ab} \mathbf{J}_{ab} \right]. \quad (155)$$

The first bracket is

$$\begin{aligned} \left[\frac{1}{2} \omega_\mu^{ab} \mathbf{J}_{ab}, \zeta^c \mathbf{G}_c \right] &= \frac{1}{2} \omega_\mu^{ab} \zeta^c [\mathbf{J}_{ab}, \mathbf{G}_c] = \frac{1}{2} \omega_\mu^{ab} \zeta^c (2\delta_{c[a} \mathbf{G}_{b]}) \\ &= \omega_\mu^{ab} \zeta^c \frac{1}{2} (\delta_{ca} \mathbf{G}_b - \delta_{cb} \mathbf{G}_a) \\ &= \frac{1}{2} \omega_\mu^{ab} \zeta_a \mathbf{G}_b - \frac{1}{2} \omega_\mu^{ab} \zeta_b \mathbf{G}_a \\ &= \frac{1}{2} (\omega_\mu^{ab} \zeta_a \mathbf{G}_b - \omega_\mu^{ba} \zeta_a \mathbf{G}_b) \quad (\text{considering that } \omega_\mu^{ab} = -\omega_\mu^{ba}) \\ &= \omega_\mu^{ab} \zeta_a \mathbf{G}_b. \end{aligned} \quad (156)$$

The second one gives us

$$\begin{aligned} \left[\omega_\mu^c \mathbf{G}_c, \frac{1}{2} \lambda^{ab} \mathbf{J}_{ab} \right] &= \frac{1}{2} \omega_\mu^c \lambda^{ab} [\mathbf{G}_c, \mathbf{J}_{ab}] = -\frac{1}{2} \omega_\mu^c \lambda^{ab} (2\delta_{c[a} \mathbf{G}_{b]}) \\ &= \frac{1}{2} (\omega_\mu^c \lambda^{ab} \delta_{ca} \mathbf{G}_b - \omega_\mu^c \lambda^{ab} \delta_{cb} \mathbf{G}_a) \\ &= \omega_\mu^c \lambda_c^a \mathbf{G}_a, \end{aligned} \quad (157)$$

and from here we read the transformation law of ω_μ^a as

$$\delta\omega_\mu^a = \partial_\mu \zeta^a - \omega_\mu^{ab} \zeta_b + \omega_\mu^b \lambda_b^a, \quad (158)$$

where we have changed the indices $a \leftrightarrow b$ in the second term to factorize $\mathbf{G}_a$ from (156). The next contribution are all the terms proportional to $\mathbf{P}_a$ and they are coming from

$$\left[e_\mu^a \mathbf{P}_a + \frac{1}{2} \omega_\mu^{ab} \mathbf{J}_{ab}, \lambda^b \mathbf{P}_b + \frac{1}{2} \lambda^{bc} \mathbf{J}_{bc} \right] = \left[e_\mu^c \mathbf{P}_c, \frac{1}{2} \lambda^{ab} \mathbf{J}_{ab} \right] + \left[\frac{1}{2} \omega_\mu^{ab} \mathbf{J}_{ab}, \lambda^c \mathbf{P}_c \right]. \quad (159)$$

The first contribution is

$$\begin{aligned} \left[\frac{1}{2} \omega_\mu^{ab} \mathbf{J}_{ab}, \lambda^c \mathbf{P}_c \right] &= \frac{1}{2} \omega_\mu^{ab} \lambda^c [\mathbf{J}_{ab}, \mathbf{P}_c] = \frac{1}{2} \omega_\mu^{ab} \lambda^c (2\delta_{c[a} \mathbf{P}_{b]}) \\ &= \frac{1}{2} \omega_\mu^{ab} \lambda^c (\delta_{ca} \mathbf{P}_b - \delta_{cb} \mathbf{P}_a) \\ &= \frac{1}{2} (\omega_\mu^{ab} \lambda_a \mathbf{P}_b - \omega_\mu^{ab} \lambda_b \mathbf{P}_a) \\ &= -\omega_\mu^{ab} \lambda_b \mathbf{P}_a. \end{aligned} \quad (160)$$

The second bracket is

$$\begin{aligned}
\left[e_\mu^c \mathbf{P}_c, \frac{1}{2} \lambda^{ab} \mathbf{J}_{ab} \right] &= \frac{1}{2} e_\mu^c \lambda^{ab} [\mathbf{P}_c, \mathbf{J}_{ab}] = -\frac{1}{2} e_\mu^c \lambda^{ab} (2\delta_{c[a} \mathbf{P}_{b]}) \\
&= -e_\mu^c \lambda^{ab} (\delta_{ca} \mathbf{P}_b - \delta_{cb} \mathbf{P}_a) \\
&= e_\mu^c \lambda_c^a \mathbf{P}_a,
\end{aligned} \tag{161}$$

and then, the transformation is given by

$$\delta e_\mu^a = \partial_\mu \lambda^a - \omega_\mu^{ab} \lambda_b + e_\mu^b \lambda_b^a. \tag{162}$$

Finally we look for contributions that are proportional to $\mathbf{J}_{ab}$. They are coming from

$$\begin{aligned}
\left[\frac{1}{2} \omega_\mu^{ab} \mathbf{J}_{ab}, \frac{1}{2} \lambda^{cd} \mathbf{J}_{cd} \right] &= \frac{1}{4} \omega_\mu^{ab} \lambda^{cd} [\mathbf{J}_{ab}, \mathbf{J}_{cd}] = \frac{1}{4} \omega_\mu^{ab} \lambda^{cd} (4\delta_{[a[d} \mathbf{J}_{c]b]}) \\
&= \frac{1}{4} \omega_\mu^{ab} \lambda^{cd} (\delta_{ad} \mathbf{J}_{cb} - \delta_{ac} \mathbf{J}_{db} - \delta_{bd} \mathbf{J}_{ca} + \delta_{bc} \mathbf{J}_{da}) \\
&= \frac{1}{4} \omega_\mu^{ab} (\lambda_a^c \mathbf{J}_{cb} + \lambda_a^d \mathbf{J}_{db} - \lambda_b^c \mathbf{J}_{ca} - \lambda_b^d \mathbf{J}_{da}) \\
&= \frac{1}{2} \omega_\mu^{ab} (\lambda_a^c \mathbf{J}_{cb} + \lambda_a^d \mathbf{J}_{db}) \\
&= \omega_\mu^{ab} \lambda_a^c \mathbf{J}_{cb} = \frac{1}{2} (\omega_\mu^{ab} \lambda_a^c \mathbf{J}_{cb} + \omega_\mu^{ab} \lambda_a^c \mathbf{J}_{cb}) \\
&= \frac{1}{2} (\omega_\mu^{ab} \lambda_a^c \mathbf{J}_{cb} + \omega_\mu^{ac} \lambda_a^b \mathbf{J}_{bc}) \\
&= \frac{1}{2} (\omega_\mu^{ab} \lambda_a^c - \omega_\mu^{ac} \lambda_a^b) \mathbf{J}_{cb}.
\end{aligned} \tag{163}$$

In the last step we just show the explicit anti-symmetry of the generator in order to have the real contributions to this expressions. Now we write the transformation as

$$\delta \omega_\mu^{ab} = \partial_\mu \lambda^{ab} + (\omega_\mu^{cb} \lambda_c^a - \omega_\mu^{ca} \lambda_c^b). \tag{164}$$

Taking all into one expression, the transformation law for the gauge field in term of each component is given by

$$\delta \tau_\mu = \partial_\mu \zeta + 2e_\mu^a \zeta_a - 2\omega_\mu^a \lambda_a, \tag{165}$$

$$\delta \omega_\mu^a = \partial_\mu \zeta^a - \omega_\mu^{ab} \zeta_b + \omega_\mu^b \lambda_b^a, \tag{166}$$

$$\delta e_\mu^a = \partial_\mu \lambda^a - \omega_\mu^{ab} \lambda_b + e_\mu^b \lambda_b^a, \tag{167}$$

$$\delta \omega_\mu^{ab} = \partial_\mu \lambda^{ab} + (\omega_\mu^{cb} \lambda_c^a - \omega_\mu^{ca} \lambda_c^b). \tag{168}$$

Also we can define the covariant derivative valued in the group and obtain the expressions

$$\delta \tau_\mu = D_\mu \zeta + e_\mu^a \zeta_a, \tag{169}$$

$$\delta \omega_\mu^a = D_\mu \zeta^a + \omega_\mu^b \lambda_b^a, \tag{170}$$

$$\delta e_\mu^a = D_\mu \lambda^a + e_\mu^b \lambda_b^a, \tag{171}$$

$$\delta \omega_\mu^{ab} = D_\mu \lambda^{ab}. \tag{172}$$

5.2. Section b) Gauge curvature components

We can directly compute the curvature components by the gauge field (147) and the definition

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]. \quad (173)$$

Starting with the derivative contributions, we see that

$$\partial_\mu A_\nu = \partial_\mu \tau_\nu \mathbf{H} + \partial_\mu \omega_\nu^a \mathbf{G}_a + \partial_\mu e_\nu^a \mathbf{P}_a + \frac{1}{2} \partial_\mu \omega_\nu^{ab} \mathbf{J}_{ab}, \quad (174)$$

then we can directly write the anti symmetric derivative as

$$\partial_{[\mu} A_{\nu]} = 2\partial_{[\mu} \tau_{\nu]} \mathbf{H} + 2\partial_{[\mu} \omega_{\nu]}^a \mathbf{G}_a + 2\partial_{[\mu} e_{\nu]}^a \mathbf{P}_a + \partial_{[\mu} \omega_{\nu]}^{ab} \mathbf{J}_{ab}. \quad (175)$$

Now, the commutator part will be calculated taking separately contributions to each generator reading the Lie algebra as it was done in the incise a). For the contributions that gives us H generators, they are coming from

$$\begin{aligned} [\omega_\mu^a \mathbf{G}_a + e_\mu^a \mathbf{P}_a, \omega_\nu^b \mathbf{G}_b + e_\nu^b \mathbf{P}_b] &= \omega_\mu^a e_\nu^b [\mathbf{G}_a, \mathbf{P}_b] + \omega_\nu^b e_\mu^a [\mathbf{P}_a, \mathbf{G}_b] \\ &= -2\omega_\mu^a e_{\nu a} \mathbf{H} + 2\omega_\nu^b e_{\mu b} \mathbf{H} \\ &= -2(\omega_{[\mu}^a e_{\nu]a}) \mathbf{H}. \end{aligned} \quad (176)$$

Now we consider the contributions that goes proportional to the generators $\mathbf{G}_a$, they come from

$$\begin{aligned} [\omega_\mu^a \mathbf{G}_a + \frac{1}{2} \omega_\mu^{ab} \mathbf{J}_{ab}, \omega_\nu^b \mathbf{G}_b + \frac{1}{2} \omega_\nu^{bc} \mathbf{J}_{bc}] &= [\omega_\mu^a \mathbf{G}_a, \frac{1}{2} \omega_\nu^{bc} \mathbf{J}_{bc}] + [\frac{1}{2} \omega_\mu^{ab} \mathbf{J}_{ab}, \omega_\nu^c \mathbf{G}_c] \\ &= \frac{1}{2} \omega_\mu^a \omega_\nu^{bc} [\mathbf{G}_a, \mathbf{J}_{bc}] + \frac{1}{2} \omega_\mu^{ab} \omega_\nu^c [\mathbf{J}_{ab}, \mathbf{G}_c] \\ &= \frac{1}{2} \omega_\mu^a \omega_\nu^{bc} (-2\delta_{a[b} \mathbf{G}_{c]}) + \frac{1}{2} \omega_\mu^{ab} \omega_\nu^c (2\delta_{c[a} \mathbf{G}_{b]}) \\ &= 2\omega_{[\mu}^a \omega_{\nu]a}^b \mathbf{G}_b. \end{aligned} \quad (177)$$

The terms proportional to $\mathbf{P}_a$ comes from

$$\begin{aligned} [e_\mu^a \mathbf{P}_a + \frac{1}{2} \omega_\mu^{ab} \mathbf{J}_{ab}, e_\nu^b \mathbf{P}_b + \frac{1}{2} \omega_\nu^{cd} \mathbf{J}_{cd}] &= \frac{1}{2} e_\mu^a \omega_\nu^{cd} [\mathbf{P}_a, \mathbf{J}_{cd}] + \frac{1}{2} e_\nu^b \omega_\mu^{ab} [\mathbf{J}_{ab}, \mathbf{P}_c] \\ &= \frac{1}{2} e_\mu^a \omega_\nu^{cd} (-2\delta_{a[c} \mathbf{P}_{d]}) + \frac{1}{2} e_\nu^b \omega_\mu^{ab} (2\delta_{c[a} \mathbf{P}_{b]}) \\ &= -e_\mu^a \omega_\nu^{bc} (\delta_{ab} \mathbf{P}_c - \delta_{ac} \mathbf{P}_b) + e_\nu^c \omega_\mu^{ab} (\delta_{ca} \mathbf{P}_b - \delta_{cb} \mathbf{P}_a) \\ &= 2e_{[\mu}^a \omega_{\nu]a}^b \mathbf{P}_b. \end{aligned} \quad (178)$$

The last contribution is the proportional to $\mathbf{J}_{ab}$, given by only one bracket

$$\begin{aligned} [\frac{1}{2} \omega_\mu^{ab} \mathbf{J}_{ab}, \frac{1}{2} \omega_\nu^{cd} \mathbf{J}_{cd}] &= \frac{1}{4} \omega_\mu^{ab} \omega_\nu^{cd} [\mathbf{J}_{ab}, \mathbf{J}_{cd}] = \frac{1}{4} \omega_\mu^{ab} \omega_\nu^{cd} (4\delta_{[a[d} \mathbf{J}_{c]b]}) \\ &= \omega_\mu^{ab} \omega_\nu^{cd} (\delta_{ad} \mathbf{J}_{cb} - \delta_{ac} \mathbf{J}_{db} - \delta_{bd} \mathbf{J}_{ca} + \delta_{bc} \mathbf{J}_{da}) \\ &= \omega_\mu^{ab} (2\omega_{\nu a}^c \mathbf{J}_{cb} - 2\omega_{\nu b}^c \mathbf{J}_{ca}) \\ &= -4\omega_\mu^{ab} \omega_{\nu a}^c \mathbf{J}_{cb} = -2(\omega_\mu^{ab} \omega_{\nu a}^c \mathbf{J}_{cb} - \omega_\mu^{ac} \omega_{\nu a}^b \mathbf{J}_{cb}) \\ &= -2(\omega_\mu^{ab} \omega_{\nu a}^c - \omega_\nu^{ab} \omega_{\mu a}^c) \mathbf{J}_{cb} \\ &= -4\omega_{[\mu}^{ab} \omega_{\nu]a}^c \mathbf{J}_{bc}. \end{aligned} \quad (179)$$

Then the components of the curvature, factorized by each generator of the algebra are ⁵

$$F_{\mu\nu}(\mathbf{H}) = 2\partial_{[\mu}\tau_{\nu]} - 4\omega_{[\mu}^a e_{\nu]a} \quad (180)$$

$$F_{\mu\nu}^a(\mathbf{G}_a) = 2\partial_{[\mu}\omega_{\nu]}^a - 2\omega_{[\mu}^{ab}\omega_{\nu]}^a \quad (181)$$

$$F_{\mu\nu}^a(\mathbf{P}_a) = 2\partial_{[\mu}e_{\nu]}^a - 2\omega_{[\mu}^{ab}e_{\nu]}^b \quad (182)$$

$$F_{\mu\nu}^{ab}(\mathbf{J}_{ab}) = 2\partial_{[\mu}\omega_{\nu]}^{ab} - 2\omega_{[\mu}^a\omega_{\nu]}^{cb} \quad (183)$$

5.3. Section c) Infinitesimal transformation of the gauge curvature

We know that the gauge curvature transforms in the adjoint representation [section C](#)

$$\delta F_{\mu\nu} = [F_{\mu\nu}, \epsilon] \quad (184)$$

so we make use of this expression for the components that we just have recover above. In this notation we can write the curvature valued in the algebra generators as

$$F_{\mu\nu} = F_{\mu\nu}(H)\mathbf{H} + F_{\mu\nu}^a(G_a)\mathbf{G}_a + F_{\mu\nu}^a(P_a)\mathbf{P}_a + \frac{1}{2}F_{\mu\nu}^{ab}(J_{ab})\mathbf{J}_{ab} \quad (185)$$

considering the definitions (180), (181), (182) and (183). The infinitesimal gauge transformation is given by

$$\delta F_{\mu\nu} = \left[F_{\mu\nu}(H)H + F_{\mu\nu}^a(G)\mathbf{G}_a + F_{\mu\nu}^a(P)\mathbf{P}_a + \frac{1}{2}F_{\mu\nu}^{ab}(J)\mathbf{J}_{ab}, \zeta\mathbf{H} + \zeta^a\mathbf{G}_a + \lambda^a\mathbf{P}_a + \frac{1}{2}\lambda^{ab}\mathbf{J}_{ab} \right]. \quad (186)$$

Following the same procedure as before, we compute first the contributions proportional to $\mathbf{H}$, reading the algebra are coming from

$$\begin{aligned} \left[F_{\mu\nu}^a(G)\mathbf{G}_a + F_{\mu\nu}^a(P)\mathbf{P}_a, \zeta^b\mathbf{G}_b + \lambda^b\mathbf{P}_b \right] &= \left[F_{\mu\nu}^a(G)\mathbf{G}_a, \lambda^b\mathbf{P}_b \right] + \left[F_{\mu\nu}^a(P)\mathbf{P}_a, \zeta^b\mathbf{G}_b \right] \\ &= F_{\mu\nu}^a(G)\lambda^b [\mathbf{G}_a, \mathbf{P}_b] + F_{\mu\nu}^a(P)\zeta^b [\mathbf{P}_a, \mathbf{G}_b] \\ &= 2 \left(F_{\mu\nu}^a(P)\zeta_a - F_{\mu\nu}^a(G)\lambda_a \right) \mathbf{H}. \end{aligned} \quad (187)$$

The contributions proportional to the generator G_a are given by

$$\begin{aligned} \left[F_{\mu\nu}^a(G)\mathbf{G}_a + \frac{1}{2}F_{\mu\nu}^{ab}(J)\mathbf{J}_{ab}, \zeta^b\mathbf{G}_b + \frac{1}{2}\lambda^{bc}\mathbf{J}_{bc} \right] &= \left[F_{\mu\nu}^a(G)\mathbf{G}_a, \frac{1}{2}\lambda^{bc}\mathbf{J}_{bc} \right] + \left[\frac{1}{2}F_{\mu\nu}^{ab}(J)\mathbf{J}_{ab}, \zeta^c\mathbf{G}_c \right] \\ &= \frac{1}{2}F_{\mu\nu}^a(G)\lambda^{bc} [\mathbf{G}_a, \mathbf{J}_{bc}] + \frac{1}{2}F_{\mu\nu}^{ab}(J)\zeta^c [\mathbf{J}_{ab}, \mathbf{G}_c] \\ &= -F_{\mu\nu}^a(G)\lambda^{bc} \left(\delta_{a[b}G_{c]} \right) + F_{\mu\nu}^{ab}(J)\zeta^c \left(\delta_{c[a}G_{b]} \right) \\ &= \frac{1}{2}F_{\mu\nu}^{ab}(J)\zeta^c (\delta_{ca}G_b - \delta_{cb}G_a) - \frac{1}{2}F_{\mu\nu}^a(G)\lambda^{bc} (\delta_{ab}G_c - \delta_{ac}G_b) \\ &= \left(F_{\mu\nu}^a(G)\lambda_a^b + F_{\mu\nu}^{ab}(J)\zeta_a \right) \mathbf{G}_b. \end{aligned} \quad (188)$$

⁵Here some signs have changed because in the results we want to factorize the generators with indices a or ab then with some renames of indices we have this resulting curvature components.

The terms that contributes with the generator $\mathbf{P}_a$ are obtained in the same way

$$\begin{aligned}
\left[F_{\mu\nu}^a(P) \mathbf{P}_a + \frac{1}{2} F_{\mu\nu}^{ab}(J) \mathbf{J}_{ab}, \lambda^c \mathbf{P}_c + \frac{1}{2} \lambda^{bc} \mathbf{J}_{bc} \right] &= \left[F_{\mu\nu}^a(P) \mathbf{P}_a, \frac{1}{2} \lambda^{bc} \mathbf{J}_{bc} \right] + \left[\frac{1}{2} F_{\mu\nu}^{ab}(J) \mathbf{J}_{ab}, \lambda^c \mathbf{P}_c \right] \\
&= \frac{1}{2} F_{\mu\nu}^a(P) \lambda^{bc} [\mathbf{P}_a, \mathbf{J}_{bc}] + \frac{1}{2} F_{\mu\nu}^{ab}(J) \lambda^c [\mathbf{J}_{ab}, \mathbf{P}_c] \\
&= -F_{\mu\nu}^a(P) \lambda^{bc} (\delta_{a[b} P_{c]}) + F_{\mu\nu}^{ab}(J) \lambda^c (\delta_{c[a} P_{b]}) \\
&= \frac{1}{2} F_{\mu\nu}^{ab}(J) \lambda^c (\delta_{ca} P_b - \delta_{cb} P_a) - \frac{1}{2} F_{\mu\nu}^a(P) \lambda^{bc} (\delta_{ab} P_c - \delta_{ac} P_b) \\
&= (F_{\mu\nu}^a(P) \lambda_a^b + F_{\mu\nu}^{ab}(J) \lambda_a) \mathbf{P}_b. \tag{189}
\end{aligned}$$

The last calculation are the contributions that are proportional to the generator $\mathbf{J}_{ab}$

$$\begin{aligned}
\left[\frac{1}{2} F_{\mu\nu}^{ab}(J) \mathbf{J}_{ab}, \frac{1}{2} \lambda^{cd} \mathbf{J}_{cd} \right] &= \frac{1}{4} F_{\mu\nu}^{ab}(J) \lambda^{cd} [\mathbf{J}_{ab}, \mathbf{J}_{cd}] = \frac{1}{4} F_{\mu\nu}^{ab}(J) \lambda^{cd} (4\delta_{[a[d} J_{c]b]}) \\
&= \frac{1}{4} F_{\mu\nu}^{ab}(J) \lambda^{cd} (\delta_{ad} \mathbf{J}_{cb} - \delta_{ac} \mathbf{J}_{db} - \delta_{bd} \mathbf{J}_{ca} + \delta_{bc} \mathbf{J}_{da}) \\
&= \frac{1}{2} F_{\mu\nu}^{ab}(J) (\lambda_a^c \mathbf{J}_{cb} - \lambda_b^c \mathbf{J}_{ca}) \\
&= \frac{1}{2} (F_{\mu\nu}^{ab}(J) \lambda_a^c \mathbf{J}_{cb} - F_{\mu\nu}^{ba}(J) \lambda_a^c \mathbf{J}_{cb}) \\
&= F_{\mu\nu}^{ab}(J) \lambda_a^c \mathbf{J}_{cb}, \tag{190}
\end{aligned}$$

and here we can make explicit the anti-symmetry of the generator

$$\begin{aligned}
F_{\mu\nu}^{ab}(J) \lambda_a^c \mathbf{J}_{cb} &= \frac{1}{2} (F_{\mu\nu}^{ab}(J) \lambda_a^c \mathbf{J}_{cb} - F_{\mu\nu}^{ac}(J) \lambda_a^b \mathbf{J}_{cb}) \\
&= \frac{1}{2} (F_{\mu\nu}^{ab}(J) \lambda_a^c - F_{\mu\nu}^{ac}(J) \lambda_a^b) \mathbf{J}_{cb} \\
&= \frac{1}{2} (F_{\mu\nu}^{ac}(J) \lambda_c^b - F_{\mu\nu}^{bc}(J) \lambda_c^a) \mathbf{J}_{ab}. \tag{191}
\end{aligned}$$

Finally we can group all the transformations in this way

$$\delta F_{\mu\nu}(H) = 2 (F_{\mu\nu}^a(P) \zeta_a - F_{\mu\nu}^a(G) \lambda_a), \tag{192}$$

$$\delta F_{\mu\nu}^a(G_a) = F_{\mu\nu}^b(G) \lambda_b^a + F_{\mu\nu}^{ba}(J) \zeta_b, \tag{193}$$

$$\delta F_{\mu\nu}^a(P) = F_{\mu\nu}^b(P) \lambda_b^a + F_{\mu\nu}^{ba}(J) \lambda_b, \tag{194}$$

$$\delta F_{\mu\nu}^{ab}(J) = \frac{1}{2} (F_{\mu\nu}^{ac}(J) \lambda_c^b - F_{\mu\nu}^{bc}(J) \lambda_c^a). \tag{195}$$

If we consider only the spatial rotations and Carroll boost, i.e., consider zero the parameters ζ and λ^a , the transformation laws are given by the same expression as [5]

$$\delta F_{\mu\nu}(H) = 2 F_{\mu\nu}^a(P) \zeta_a, \tag{196}$$

$$\delta F_{\mu\nu}^a(G) = F_{\mu\nu}^b(G) \lambda_b^a - F_{\mu\nu}^{ab}(J) \zeta_b, \tag{197}$$

$$\delta F_{\mu\nu}^a(P) = F_{\mu\nu}^b(P) \lambda_b^a, \tag{198}$$

$$\delta F_{\mu\nu}^{ab}(J) = \frac{1}{2} (F_{\mu\nu}^{ac}(J) \lambda_c^b - F_{\mu\nu}^{bc}(J) \lambda_c^a). \tag{199}$$

Comparing with [5], they have a change in the definition (147) have changed $\lambda^a \leftrightarrow \zeta^a$, so the result is consistent.

Appendix A. Stokes theorem and the boundary terms

In this appendix we state the Stokes theorem for differential forms and justify why the total derivative terms appearing in the variation of the Maxwell and Yang-Mills actions can be discarded.

The theorem

Let M be an orientable d -dimensional manifold with boundary ∂M , and let $\omega \in \Omega^{d-1}(M)$ be an *ordinary* (real-valued) $(d-1)$ -form. The Stokes theorem states that

$$\int_M d\omega = \int_{\partial M} \omega. \quad (200)$$

The degrees match on both sides: $d\omega$ is a top form on the d -dimensional bulk M , while ω is a top form on the $(d-1)$ -dimensional boundary ∂M . This single statement contains all the classical integral theorems of vector calculus: the fundamental theorem of calculus ($d=1$), the Stokes circulation theorem ($d=2$) and the Gauss divergence theorem ($d=3$) are the cases obtained for the corresponding dimensions and form degrees.

The boundary term in Maxwell theory

In the variation of the Maxwell action the following term appears

$$\int_M d(\delta A \wedge \star F). \quad (201)$$

The object $\omega = \delta A \wedge \star F$ is the wedge product of a 1-form with a 2-form, so $\omega \in \Omega^3(M)$ is an ordinary 3-form in $d=4$: exactly a $(d-1)$ -form, and both requirements above are satisfied directly because the abelian theory has no algebra factor. Applying (200)

$$\int_M d(\delta A \wedge \star F) = \int_{\partial M} \delta A \wedge \star F. \quad (202)$$

The variational principle is defined by comparing field configurations with the same values on the boundary, that is, the one-parameter family $A_s = A + s \delta A$ must coincide with A on ∂M , which imposes

$$\delta A|_{\partial M} = 0. \quad (203)$$

Since the integrand is linear in δA , it vanishes pointwise on ∂M and therefore

$$\int_{\partial M} \delta A \wedge \star F = 0. \quad (204)$$

If M has no boundary the integral vanishes trivially, and if $M = \mathbb{R}^4$ the same conclusion follows by demanding sufficiently fast fall-off of the fields at infinity.

The boundary term in Yang-Mills theory

In the non-abelian case the term appearing in the variation is

$$\int_M d \operatorname{Tr}(\delta A \wedge \star F), \quad (205)$$

and here the second requirement becomes non trivial: the object $\delta A \wedge \star F \in \Omega^3(M) \otimes \mathfrak{g}$ is valued in the Lie algebra, and the Stokes theorem does not apply to matrix-valued forms.

This is precisely the role of the trace. Since $\text{Tr} = \text{id}_{\Omega^p} \otimes \text{tr}_{\mathfrak{g}}$ acts as the identity on the form factor and reduces the algebra factor to a number,

$$\omega = \text{Tr}(\delta A \wedge \star F) = \delta A^A \wedge \star F^B \text{tr}(T_A T_B) = \delta A^A \wedge \star F^A \in \Omega^3(M) \quad (206)$$

is an ordinary 3-form, and the theorem applies. Moreover, the exterior derivative commutes with the trace,

$$d \text{Tr}(\Phi) = \text{Tr}(d\Phi), \quad (207)$$

because the trace is an algebraic operation with constant coefficients, $\text{tr}(T_A T_B) = \delta_{AB}$, which the derivative does not see. This is what allows us to first form the ordinary 3-form with the trace and then recognize the integrand as its exact derivative. Applying (200) and the boundary condition (203)

$$\int_M d \text{Tr}(\delta A \wedge \star F) = \int_{\partial M} \text{Tr}(\delta A \wedge \star F) = 0. \quad (208)$$

In both theories the conclusion is the same: the total derivative produced by the integration by parts is the bulk integral of an exact ordinary form, the Stokes theorem converts it into a boundary integral, and the boundary conditions of the variational principle make it vanish. The only structural difference is that in the non-abelian case the trace must be taken *before* invoking the theorem, in order to have an ordinary form to which it applies.

Appendix B. Curvature in the coordinate basis

The definition $F = dA + A \wedge A$ is built solely out of the exterior derivative and the wedge product, both of which are independent of any choice of basis for $\Omega^\bullet(M)$. To make this explicit, we repeat the computation in the coordinate basis $\{dx^\mu\}$ and check that it reproduces the same gauge curvature. The coframe and the coordinate basis are related by $e^a = e^a_\mu dx^\mu$, with e^a_μ the vierbein, so that the algebra-valued connection reads

$$A = A_a^C e^a \otimes T_C = A_a^C e^a_\mu dx^\mu \otimes T_C \equiv A_\mu^C dx^\mu \otimes T_C, \quad (209)$$

where we defined the coordinate components $A_\mu^C := e^a_\mu A_a^C$. Since the coordinate basis is holonomic (it is an assumption here), $d(dx^\nu) = 0$ exactly, and the first contribution follows directly from the exterior derivative

$$\begin{aligned} dA &= d(A_\nu^C dx^\nu \otimes T_C) = \partial_\mu A_\nu^C dx^\mu \wedge dx^\nu \otimes T_C \\ &= \frac{1}{2} (\partial_\mu A_\nu^C - \partial_\nu A_\mu^C) dx^\mu \wedge dx^\nu \otimes T_C \\ &= \partial_{[\mu} A_{\nu]}^C dx^\mu \wedge dx^\nu \otimes T_C. \end{aligned} \quad (210)$$

The next contribution is obtained by replacing the same definition

$$\begin{aligned} A \wedge A &= (A_\mu^A dx^\mu \otimes T_A) \wedge (A_\nu^B dx^\nu \otimes T_B) = (A_\mu^A A_\nu^B dx^\mu \wedge dx^\nu) \otimes T_A T_B \\ &= \frac{1}{2} (A_\mu^A A_\nu^B dx^\mu \wedge dx^\nu \otimes T_A T_B - A_\nu^A A_\mu^B dx^\mu \wedge dx^\nu \otimes T_A T_B) \\ &= \frac{1}{2} A_\mu^A A_\nu^B dx^\mu \wedge dx^\nu \otimes T_A T_B - \frac{1}{2} A_\nu^B A_\mu^A dx^\mu \wedge dx^\nu \otimes T_B T_A \\ &= \frac{1}{2} A_\mu^A A_\nu^B dx^\mu \wedge dx^\nu \otimes (T_A T_B - T_B T_A), \end{aligned} \quad (211)$$

and using the Lie algebra $[T_A, T_B] = f_{AB}^C T_C$ we have

$$\begin{aligned} A \wedge A &= \frac{1}{2} A_\mu^A A_\nu^B dx^\mu \wedge dx^\nu \otimes (T_A T_B - T_B T_A) \\ &= \frac{1}{2} A_\mu^A A_\nu^B f_{AB}^C dx^\mu \wedge dx^\nu \otimes T_C. \end{aligned} \quad (212)$$

Adding both contributions, the curvature in the coordinate basis is

$$\begin{aligned} F &= dA + A \wedge A \\ &= \partial_{[\mu} A_{\nu]}^C dx^\mu \wedge dx^\nu \otimes T_C + \frac{1}{2} A_\mu^A A_\nu^B f_{AB}^C dx^\mu \wedge dx^\nu \otimes T_C \\ &= \frac{1}{2} \left(\partial_\mu A_\nu^C - \partial_\nu A_\mu^C + A_\mu^A A_\nu^B f_{AB}^C \right) dx^\mu \wedge dx^\nu \otimes T_C \\ &= \frac{1}{2} F_{\mu\nu}^C dx^\mu \wedge dx^\nu \otimes T_C, \end{aligned} \quad (213)$$

with the field-strength components

$$F_{\mu\nu}^C = \partial_\mu A_\nu^C - \partial_\nu A_\mu^C + f_{AB}^C A_\mu^A A_\nu^B. \quad (214)$$

This is identical in structure to the frame-basis result F_{ab}^C under the replacement $a \rightarrow \mu$, $b \rightarrow \nu$, which confirms that the definition $F = dA + A \wedge A$ yields the same curvature in either basis.

Appendix C. Infinitesimal transformation of F

The gauge curvature associated with a Lie-algebra-valued connection A transforms in the adjoint representation of the gauge group. For a gauge transformation $g \in G$, the curvature transforms as [3]

$$F' = g F g^{-1}. \quad (215)$$

In the non-abelian case such as the Carroll algebra we can derive this transformation rule from the infinitesimal parameter and the gauge field also. Since we are considering an infinitesimal gauge transformation, the spacetime coordinates remain unchanged and the variation δ acts only on the Lie-algebra-valued fields. Starting from the definitions (146), (147) we can express δF directly

$$\delta F_{\mu\nu} = \delta (\partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]), \quad (216)$$

we can expand in the coordinate basis and use the transformation rule for the gauge field A , the first contribution is

$$\begin{aligned} \delta (\partial_\mu A_\nu - \partial_\nu A_\mu) &= \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu \\ &= \partial_\mu (\partial_\nu \epsilon + [A_\nu, \epsilon]) - \partial_\nu (\partial_\mu \epsilon + [A_\mu, \epsilon]) \\ &= \partial_\mu \partial_\nu \epsilon - \partial_\nu \partial_\mu \epsilon + [\partial_\mu A_\nu, \epsilon] + [A_\nu, \partial_\mu \epsilon] - [\partial_\nu A_\mu, \epsilon] - [A_\mu, \partial_\nu \epsilon] \\ &= [\partial_\mu A_\nu - \partial_\nu A_\mu, \epsilon] + [A_\nu, \partial_\mu \epsilon] - [A_\mu, \partial_\nu \epsilon] \\ &= [\partial_{[\mu} A_{\nu]}, \epsilon] + [A_\nu, \partial_\mu \epsilon] - [A_\mu, \partial_\nu \epsilon]. \end{aligned} \quad (217)$$

Now the second contribution is

$$\begin{aligned} \delta [A_\mu, A_\nu] &= [\delta A_\mu, A_\nu] + [A_\mu, \delta A_\nu] \\ &= [\partial_\mu \epsilon + [A_\mu, \epsilon], A_\nu] + [A_\mu, \partial_\nu \epsilon + [A_\nu, \epsilon]] \\ &= [\partial_\mu \epsilon, A_\nu] + [A_\mu, \partial_\nu \epsilon] + [[A_\mu, \epsilon], A_\nu] + [A_\mu, [A_\nu, \epsilon]] \\ &= [\partial_\mu \epsilon, A_\nu] + [A_\mu, \partial_\nu \epsilon] + [[A_\mu, A_\nu], \epsilon], \end{aligned} \quad (218)$$

where in the last step we have used the Jacobi identity and the previous steps are just properties of this bracket. Using the Jacobi identity

$$[[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] = 0, \quad (219)$$

with $X = A_\mu$, $Y = \epsilon$ and $Z = A_\nu$, one finds

$$[[A_\mu, \epsilon], A_\nu] + [A_\mu, [A_\nu, \epsilon]] = [[A_\mu, A_\nu], \epsilon]. \quad (220)$$

Now we can add all the contributions to find

$$\begin{aligned} \delta F_{\mu\nu} &= [\partial_{[\mu} A_{\nu]}, \epsilon] + [A_\nu, \partial_\mu \epsilon] - [A_\mu, \partial_\nu \epsilon] + [\partial_\mu \epsilon, A_\nu] + [A_\mu, \partial_\nu \epsilon] + [[A_\mu, A_\nu], \epsilon] \\ &= [\partial_{[\mu} A_{\nu]}, \epsilon] + [[A_\mu, A_\nu], \epsilon] \\ &= [\partial_{[\mu} A_{\nu]} + [A_\mu, A_\nu], \epsilon] \\ &= [F_{\mu\nu}, \epsilon], \end{aligned} \quad (221)$$

Therefore, the curvature transforms covariantly in the adjoint representation,

$$\delta F_{\mu\nu} = [F_{\mu\nu}, \epsilon], \quad (222)$$

which is precisely the infinitesimal version of

$$F' = g F g^{-1}. \quad (223)$$

Also this can be seen in the differential forms version

$$\delta A = d\epsilon + [A, \epsilon], \quad (224)$$

and with the definition of the gauge curvature

$$F = dA + A \wedge A. \quad (225)$$

We use the same arguments as before and obtain

$$\delta F = \delta(dA + A \wedge A) = d(\delta A) + \delta(A \wedge A). \quad (226)$$

Using the Leibniz rule for the variation of the wedge product,

$$\delta(A \wedge A) = (\delta A) \wedge A + A \wedge (\delta A), \quad (227)$$

we obtain

$$\delta F = d(\delta A) + (\delta A) \wedge A + A \wedge (\delta A). \quad (228)$$

Substituting the infinitesimal transformation of the gauge field we have

$$\begin{aligned} \delta F &= d(d\epsilon + [A, \epsilon]) + (d\epsilon + [A, \epsilon]) \wedge A + A \wedge (d\epsilon + [A, \epsilon]) \\ &= d[A, \epsilon] + d\epsilon \wedge A + A \wedge d\epsilon + [A, \epsilon] \wedge A + A \wedge [A, \epsilon], \end{aligned} \quad (229)$$

where we have used the nilpotency of the exterior derivative,

$$d^2 = 0. \quad (230)$$

The first term can be expanded using the graded Leibniz rule [1],

$$d[A, \epsilon] = [dA, \epsilon] - [A, d\epsilon], \quad (231)$$

so that

$$\begin{aligned}\delta F &= [dA, \epsilon] - [A, d\epsilon] + d\epsilon \wedge A + A \wedge d\epsilon \\ &\quad + [A, \epsilon] \wedge A + A \wedge [A, \epsilon].\end{aligned}\tag{232}$$

The terms containing derivatives of the gauge parameter cancel

$$\delta F = [dA, \epsilon] + [A, \epsilon] \wedge A + A \wedge [A, \epsilon].\tag{233}$$

Finally, using the Jacobi identity, the last two terms can be combined as

$$[A, \epsilon] \wedge A + A \wedge [A, \epsilon] = [A \wedge A, \epsilon],\tag{234}$$

and therefore

$$\begin{aligned}\delta F &= [dA, \epsilon] + [A \wedge A, \epsilon] \\ &= [dA + A \wedge A, \epsilon] \\ &= [F, \epsilon].\end{aligned}\tag{235}$$

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